

Beyond static risk aversion: Online multi-period mean-variance portfolio optimization using hidden Markov chains

Siya Liu^{1†}, Chengneng Jin^{1†}, Jianjun Gao^{1*}, Weiping Wu²

^{1*}School of Information Management and Engineering, Shanghai University of Finance and Economics, Shanghai, 200433, Shanghai, China.

²School of Economics and Management, Fuzhou University, Fuzhou, 350108, Fujian, China.

*Corresponding author(s). E-mail(s): gao.jianjun@shufe.edu.cn, ORCID: 0000-0002-7824-5919;

Contributing authors: liusiya7@stu.sufe.edu.cn, ORCID: 0009-0009-6841-2160; jin.chengneng@gmail.com, ORCID: 0009-0004-7295-2286; wu.weiping@fzu.edu.cn, ORCID: 0000-0002-5958-6910;

[†]These authors contributed equally to this work.

Abstract

Recognizing that investors' risk preferences evolve in response to dynamically changing market conditions, which are often unobservable and exhibit regime-switching behavior, this paper addresses multi-period mean-variance (MMV) portfolio optimization with market-regime-dependent risk parameters and partially observed market information. The model incorporates a regime-switching framework represented by hidden Markov models (HMM) with Gaussian emission probabilities. This approach mirrors financial practice, where continuously observable variables, such as the Volatility Index (VIX), are used as proxies for underlying market information. HMM not only provides a tractable method for updating risk aversion and model parameters in an online learning fashion, but it also integrates seamlessly into the MMV solution framework. In the decision phase, we derive closed-form solutions for the MMV model in two scenarios: one with only risky assets and another that includes a risk-free asset. The proposed policies generalize several classical policies as special cases. Using real market data, we compare our model with alternatives that either ignore the regime-switching structure or assume constant risk aversion. Our experiments

demonstrate that the proposed model consistently outperforms these benchmarks, highlighting the importance of incorporating dynamic market conditions and evolving risk preferences into practical portfolio management.

Keywords: Multi-period mean-variance portfolio decision, Regime-switching model, Hidden Markov model, Time-varying risk aversion, Dynamic asset allocation

1 Introduction

Since Markowitz [1] introduces the single-period mean-variance (MV) portfolio selection model, it has been significantly extended in both academic research and practical applications [2]. One major advancement is the development of the multi-period mean-variance (MMV) portfolio optimization model. Given that variance is not separable in the context of dynamic programming, Li and Ng [3] propose the embedding method, yielding a closed-form portfolio policy. Building on this approach, the MMV model has been explored in various settings (see the survey by Cui et al. [4]). While some studies assume that asset returns are independent over time (e.g., [3, 5]), others adopt the more realistic assumption that market states exhibit certain correlations (e.g., [6–9]). In the MV portfolio optimization, the risk aversion parameter plays a crucial role. With few exceptions¹, most studies assume risk aversion to be deterministic or follow a simple wealth-dependent structure. Although some financial literature highlights the importance of time-varying risk aversion (e.g., [12, 13]), little research has integrated regime-switching market conditions with market state-dependent risk aversion in the MMV model. This work aims to fill that gap.

The seminal work of Hamilton [14] demonstrates that market conditions are time-varying and frequently alternate between different states over time. Building on this insight, numerous studies have integrated regime-switching structures into the MMV model (e.g., [6, 7, 15]). However, in practice, the true market state is often unobservable, as highlighted by Li et al. [16]. To address this issue, we adopt a hidden Markov model (HMM) framework, which has been widely used in financial settings to capture the regime-switching behavior of market states (e.g., [17, 18]). Recognizing that observable indicators, such as the Volatility Index (VIX), are often continuous variables [17], we employ an HMM with Gaussian emission probabilities. Regarding the modeling of time-varying risk aversion, previous research indicates that risk aversion can be influenced by factors such as habits, personal circumstances, and investment performance (e.g., [12, 19]). Nevertheless, market conditions play a critical role in shaping risk aversion. For example, Guiso et al. [20] found that investors became more risk-averse after the 2008 financial crisis. Motivated by these observations, this study investigates the MMV portfolio optimization problem within a regime-switching market modeled by an HMM, incorporating risk aversion that depends on the market conditions.

This work makes several significant contributions to MMV portfolio optimization. First, we propose a practical and tractable MMV portfolio model that incorporates

¹Basak and Chabakauri [10] and Björk et al. [11] consider wealth-dependent risk aversion in a continuous-time MV portfolio model, though they assume independent asset returns.

unobservable regime-switching dynamics and market-state-dependent risk aversion. Using a hidden Markov model, where only historically observable states are assumed to be known, we sequentially learn and filter the transition probability matrix and market states. Second, we derive closed-form portfolio policies in decision phase for two key scenarios: one involving only risky assets and the other including a risk-free asset. We conquer the difficulty in using embedding method brought by the unobservable parameters. Once transition probabilities are estimated and market states are filtered, the portfolio policy can be computed by explicit formula. Such a solution subsumes several known MMV portfolio with full market information as its special cases. Notably, our approach facilitates online implementation, enabling the use of the most recent data while discarding outdated information, rather than relying on fixed parameters, thus enhancing practical applicability. Finally, using real market data, we compare our portfolio policies against several benchmark strategies, including those that ignore Markov regime-switching dynamics and assume a constant risk aversion. The numerical results demonstrate that our model consistently outperforms these benchmarks in out-of-sample tests, underscoring the advantages of incorporating regime-switching dynamics and time-varying risk aversion in stochastic financial markets. These results suggest the need for caution in building portfolio optimization models, as oversimplification may compromise performance.

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The remainder of this paper is organized as follows. Section 2 provides a review of related literature. Section 3 presents the mathematical formulation of our model. Section 4 describes the overall online implementation procedure of portfolio policies, derives the optimal investment strategies for two market settings, and discusses some

special cases and connections with existing models. Section 5 presents numerical results and some analysis. Finally, Section 6 concludes the paper and gives suggestions for future research.

Notations: We employ boldface letters to represent vectors or matrices. The notation $\mathbf{1}$ denotes the all-one vector of appropriate dimension. We use $\mathbf{B} \succ 0$ to represent that $\mathbf{B}$ is a positive definite matrix. We use $\mathcal{N}(\mu, \Sigma)$ to represent the multivariate normal distribution where μ and Σ are the mean and covariance, respectively.

2 Related Work

Following the seminal work of Markowitz [1], the single-period mean-variance optimization model has been extended in various directions, including, but not limited to, [2, 21, 22]. Among these extensions, multi-period models are particularly advantageous, as they consider long-term potential benefits and help mitigate myopic decision-making (see [23, 24]). This paper focuses on an asset allocation framework based on the multi-period formulation.

One important approach to solving MMV problems is the embedding method, which involves solving an auxiliary parameterized linear-quadratic (LQ) problem first and then retrieving the solution by identifying the appropriate parameter. The policy obtained through this method is referred to as a *pre-commitment* policy in the literature (see [4, 10]). For instance, Gao and Li [25] considered generalized correlations between asset returns, Cui et al. [5] imposed a no-shorting constraint, Gao et al. [8] limited the number of investment periods for risky assets, and Wu et al. [26] solved the dynamic MV portfolio optimization problem with general linear constraints. An alternative approach, termed the *time-consistent* approach, was proposed by Basak and Chabakauri [10]. This method reformulates the problem as a sequential intrapersonal game and solves it by adopting the subgame Nash equilibrium policy. Comparative studies by van Staden et al. [27, 28] have demonstrated the superior performance of the pre-commitment strategy in terms of terminal wealth distribution and robustness to model misspecification. Given these advantages, we adopt the pre-commitment strategy to solve the dynamic MV optimization problem with regime-dependent risk aversion.

Risk aversion, an investor's preference for certainty, plays an important role in many decision-making problems [29, 30]. In the past few years, economists have conducted various experiments to test the stability of risk attitude and find that risk profile can be influenced by market states [20, 31, 32] and other factors like personal conditions, consumption and investment performance [33, 34]. To incorporate dynamic risk aversion, Li et al. [16] adopted a market dependent risk aversion in a utility function, finding that the optimal policy can react to changing market conditions. As for the MV model, Björk et al. [11] modeled the risk aversion parameter as being inversely proportional to wealth and solved the problem analytically using the time-consistent control theory. Wu [35] studied the time-consistent policies of dynamic MV problem in a discrete-time setting, and derived the analytical solutions with constant or wealth-dependent risk aversion parameters, respectively. Wu and Chen [15] considered that not only are the expected returns of risky assets state-dependent, but

also the risk aversion depends on time-varying macroeconomic market conditions and personal information, where the states follow a fixed Markov chain.

Regime-switching, introduced by Hamilton [14], is used to describe abrupt changes in business cycles and was later extended to model transitions between economic growth and financial distress. It has since been employed in developing various portfolio strategies [8, 9, 36, 37]. In the context of the MMV formulation, Wu and Li [38] and Wu et al. [39] study the problem under regime-switching settings with stochastic cash flows and state-dependent exit probabilities, respectively. Wu et al. [40] explored an investment–consumption optimization problem with a time-varying utility function in a Markov regime-switching market. However, these studies mainly assume constant risk aversion and fully observable Markov chains to describe the regime-switching process, overlooking the unobservable state, which is more representative of financial practice. One notable exception is Zhang et al. [18], which combines hidden Markov models with the MMV portfolio selection framework and solves for a pre-committed policy. Our work differs from Zhang et al. [18] in several key aspects. Unlike their assumption of constant risk aversion, we incorporate market-state-dependent risk aversion, which aligns more closely with empirical findings and introduces additional challenges in developing a solution. Moreover, we propose an online learning framework that integrates the estimation and portfolio decision phases, enhancing practical implementation. More importantly, in the portfolio decision phase, we provide an explicit solution for the MMV model that involves only risky assets, which offers a more general formulation compared to models that incorporate both risky and risk-free assets.²

To the best of our knowledge, this is the first study to incorporate both regime-switching market conditions represented by a hidden Markov model and market-state-dependent risk aversion in an MMV portfolio optimization framework and solve it by pre-commitment approach.

3 Model Framework

In this section, we present the mathematical formulation of our proposed model. We begin by developing a regime-switching framework based on the hidden Markov model to capture the dynamics of unobservable market states in Section 3.1. Subsequently, in Section 3.2, we introduce the MMV portfolio optimization problem, incorporating time-varying risk aversion that depends on the inferred market regimes. This formulation enables us to capture both the evolving nature of market conditions and the adaptive risk preferences of investors within a unified framework.

For notations, let $(\Omega, \{\mathcal{F}_t\}_{t=0}^T, \mathbb{P})$ be a complete probability space, where $\mathbb{P}$ denotes the probability measure and $\{\mathcal{F}_t\}_{t=0}^T$ represents the filtration satisfying $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$. All random variables in this study are defined on this probability space. We denote the conditional expectation and variance of a random variable X given $\mathcal{F}_t$ as $E_t[X] = E[X|\mathcal{F}_t]$ and $\text{Var}_t[X] = \text{Var}[X|\mathcal{F}_t]$, respectively. The unconditional expectation and variance are denoted as $E_0[X] = E[X|\mathcal{F}_0]$ and

²A risk-free asset can be viewed as a special asset with a deterministic return rate and no correlation with other risky assets. Therefore, the solution for the model with only risky assets subsumes the case that includes both risky and risk-free assets. The reverse, however, does not hold.

$\text{Var}_0[X] = \text{Var}[X|\mathcal{F}_0]$. To emphasize the dependency on the market state s_t , we use the notation $\text{E}_t[X|s_t]$ and $\text{Var}_t[X|s_t]$ when there is no ambiguity.

3.1 HMM Based Regime-Switching Model

We examine two distinct investment scenarios within regime-switching framework, differentiated by the composition of available assets. The first scenario considers a portfolio comprised of solely risky assets, while the second incorporates both risky and risk-free assets. Although the latter scenario is more prevalent in MMV model literature [3, 26, 35, 38, 39, 41], the former also has significant practical implications [25, 42]. We derive optimal policies for both scenarios to provide comprehensive insights.

We model the dynamics of market states using a finite-state first-order Markov chain [6, 15, 38, 39]. Let $s_t \in \mathbb{M} \triangleq \{1, 2, \dots, M\}$ denote the market state at time t , where M is the total number of states. The one-step state transition probability is denoted as $\mathbb{P}(s_{t+1} = j|s_t = i)$ for all $i, j \in \mathbb{M}$ and $t = 0, 1, \dots, T-1$. We can use a $M \times M$ transition matrix $\mathbf{Q}$, where the (i, j) -entry $Q(i, j)$ in the transition matrix $\mathbf{Q}$ is the transition probability $\mathbb{P}(s_{t+1} = j|s_t = i)$, to describe the stochastic properties of $\{s_t\}_{t=0}^T$. All elements in $\mathbf{Q}$ are nonnegative and the elements of each row sum to 1:

$$\sum_{j=1}^M Q(i, j) = 1, \quad Q(i, j) \geq 0, \quad \forall i, j \in 1, \dots, M. \quad (1)$$

The future state distribution can be computed using the k -step transition probability, defined as $\mathbb{P}(s_{t+k} = j|s_t = i)$. If the distribution of the market state s_t is $\pi_t \triangleq (\mathbb{P}(s_t = 1), \mathbb{P}(s_t = 2), \dots, \mathbb{P}(s_t = M))$, then the distribution of the market state s_{k+t} in the future can be calculated by:

$$\pi_{k+t} = \pi_t \mathbf{Q}^k, \quad (2)$$

where $\pi_{k+t} \triangleq (\mathbb{P}(s_{k+t} = 1), \mathbb{P}(s_{k+t} = 2), \dots, \mathbb{P}(s_{k+t} = M))$ and $\mathbf{Q}^k$ is the k -th power of matrix $\mathbf{Q}$.

In the first scenario, we consider a market with n risky assets, indexed by $1, \dots, n$, traded over T discrete time periods. Let $S_t(i)$ denote the price of the i -th asset at time t . The return of asset i from time $t-1$ to t is defined as $R_t(i) = S_t(i)/S_{t-1}(i)$ for $i = 1, \dots, n$ and $t = 1, \dots, T$. The random return vector is denoted as $\mathbf{R}_t = (R_t(1), \dots, R_t(n))^\top$, which is $\mathcal{F}_t$ -measurable. Conditional on $s_t = i$, we define $\mathbf{r}_t(i) \triangleq \text{E}_t[\mathbf{R}_{t+1}|s_t = i]$ and $\Sigma_t(i) \triangleq \text{Cov}_t[\mathbf{R}_{t+1}|s_t = i] = \text{E}_t[\mathbf{R}_{t+1}\mathbf{R}_{t+1}^\top|s_t = i] - \mathbf{r}_t(i)\mathbf{r}_t(i)^\top$ as the mean and covariance matrix of risky assets' returns, respectively.

In the second scenario, we introduce a risk-free asset with a state-dependent return $r_{f,t}(i)$, where $s_t = i$ represents the market state. This aligns with empirical observations that risk-free rates, often proxied by T-bill returns [16, 27, 28], fluctuate over extended time horizons. We define the excess return vector as $\mathbf{R}_t^e = (R_t^e(1), \dots, R_t^e(n))^\top$ for $t = 1, \dots, T$. The conditional mean of excess returns is given by $\mathbf{r}_t^e(i) \triangleq \text{E}_t[\mathbf{R}_{t+1}^e|s_t = i] = \mathbf{r}_t(i) - r_{f,t}(i)\mathbf{1}$, and the conditional covariance matrix is $\Sigma_t(i) = \text{Cov}_t[\mathbf{R}_{t+1}^e|s_t = i] = \text{Cov}_t[\mathbf{R}_{t+1}|s_t = i]$.

To ensure our model is well-defined and solvable, we introduce an assumption that is generalized from a common condition of models with independent asset returns [3]: **Assumption 1.** *The covariance matrix $\Sigma_t(i)$ is positive definite for all market states $i \in \mathbb{M}$ and time periods $t = 0, \dots, T - 1$.*

This assumption guarantees that the covariance structure of asset returns remains non-degenerate across all market states and time periods, which is crucial for the mathematical tractability and economic meaningfulness of our model.

While many existing MMV studies assume that the market state at any time is fully observable and follows a fixed and known Markov chain [15, 38, 39], we consider a more realistic setting where the regime is only partially observed and some observations, such as the VIX index [16] or MSCI World index [17], are used as predictors or signals for forecasting upcoming market states. Consequently, our model employs a hidden Markov model rather than a simple Markov chain.

Let V_t denote the observation at time t . The probability distribution of stochastic process $\{V_t\}_{t=0}^T$ depends on the realization of a hidden discrete stochastic process, i.e., the market states $\{s_t\}_{t=0}^T$. While $\{V_t\}_{t=0}^T$ are directly observable, the hidden random variable s_t is only indirectly observable, and all statistical inference with respect to it must be done by inspecting its effect on the realization of V_t [43]. In this paper, we adopt a Gaussian HMM to model the probability distribution of the observations $\{V_t\}_{t=0}^T$, a more complex situation compared to the discrete distribution in [18]. For example, consider a two-state HMM where the conditional distribution of observations in each state is a Gaussian distribution:

$$V_t | s_t \sim \mathcal{N}(\mu_{s_t}, \sigma_{s_t}^2), \quad (3)$$

where

$$\mu_{s_t} = \begin{cases} \mu_1, & \text{if } s_t = 1 \\ \mu_2, & \text{if } s_t = 2 \end{cases}, \quad \sigma_{s_t} = \begin{cases} \sigma_1, & \text{if } s_t = 1 \\ \sigma_2, & \text{if } s_t = 2 \end{cases}, \quad \mathbf{Q} = \begin{bmatrix} Q(1, 1) & Q(1, 2) \\ Q(2, 1) & Q(2, 2) \end{bmatrix}.$$

This two-state model offers intuitive practical interpretations. A common application is to characterize market conditions as either bull or bear states. The bull market state can be modeled by a Gaussian distribution with higher mean and lower variance, while the bear market state is represented by a Gaussian distribution with lower mean and higher variance [17]. In our problem, all inferences with respect to the hidden market state should be based on observations V_t .

3.2 MMV Portfolio Optimization Problem

Consider an investor who enters the market at time $t = 0$ with initial wealth $x_0 > 0$ and aims to invest in a financial market over a fixed horizon of T discrete periods, from $t = 0$ to $T - 1$. Let x_t denote the wealth available for investment at time t , and $\mathbf{u}_t = (u_t(1), u_t(2), \dots, u_t(n))^\top$ represent the portfolio decision vector, where $u_t(i)$ is the amount invested in the i -th risky asset during the interval $[t, t + 1)$. It is important to note that $\mathbf{u}_t$ must be $\mathcal{F}_t$ -measurable, as it can only utilize historical information up to time t .

Under the self-financing constraint, we consider two scenarios. If there is no risk-free asset in the market, then at each re-balancing time t , the portfolio allocation should be equal to the budget, so the wealth process will evolve as

$$\begin{aligned} x_t &= \mathbf{1}^\top \mathbf{u}_t, \\ x_{t+1} &= \mathbf{R}_{t+1}^\top \mathbf{u}_t, \end{aligned} \quad (4)$$

for $t = 0, \dots, T-1$. If there is one risk-free asset available for investment and the portfolio decision at time t is $\mathbf{u}_t$, then the wealth invested in the risk-free asset is $x_t - \sum_{i=1}^n u_t(i)$, therefore, x_{t+1} can be represented by the following wealth dynamics equation

$$x_{t+1} = r_{f,t}x_t + (\mathbf{R}_{t+1}^e)^\top \mathbf{u}_t, \quad t = 0, \dots, T-1. \quad (5)$$

Besides, We adopt the following assumptions, which are common in the literature on MMV portfolio selection problems [15, 38, 39]:

Assumption 2. *In this market, (a) short selling is allowed for all assets in all periods, (b) no capital additions or withdrawals are allowed during the investment horizon, and (c) there is no transaction costs.*

Note that, by applying the methods from [15] and [26], the current solution framework can be extended to situations where certain conditions are not satisfied, such as the presence of stochastic cash flows in the wealth process or the imposition of short-selling constraints on risky assets.

Motivated by the work of Li et al. [16], this research focuses on incorporating time-varying risk aversion parameters into the MMV model. While existing literature typically assumes a constant or deterministic risk aversion parameter across all periods in the MMV model (e.g., see the survey by Cui et al. [4]), empirical studies suggest that an investor's risk profile is state-dependent [13, 33]. For instance, an investor is generally less risk-averse when the market is in favorable conditions and more risk-averse when the market is in a downturn. To model this time-varying risk profile, we introduce a $\mathcal{F}_T$ -measurable discrete random variable ω to represent the investor's risk aversion parameter, with support $\{\omega(1), \dots, \omega(M)\}$. The risk parameter depends on the market state s_T , such that $\omega = \omega(i)$ when $s_T = i$ for $i = 1, \dots, M$. That is to say, at any time $t < T$, the investor only knows the estimation of the risk parameter, i.e., we define the expected risk aversion of ω as follows,

$$\omega_t = \sum_{j \in \mathbb{M}} \mathbb{P}(s_T = j | \mathcal{F}_t) \omega(j), \quad t = 0, \dots, T-1, \quad (6)$$

where $\mathbb{P}(s_T = j | \mathcal{F}_t)$ is the probability that the market regime will be j at time T conditional on the information set $\mathcal{F}_t$ and can be computed using (2).

As the market state is unobservable, we learn the market state s_t along with the HMM parameters through the emission process, such as the observation of VIX. This learning process leads to updates of the MMV parameters, including expected risk aversion, and the expected value and covariance matrix of the assets. This approach sharply contrasts with the classical MMV model, where all parameters are estimated at time $t = 0$ using information $\mathcal{F}_0$. Our learning process is more aligned with real

financial practice, i.e., as investors acquire more information, estimates should be updated accordingly. This combination of learning and decision-making, akin to online portfolio selection [44, 45], ensures that strategies evolve as new data arrives. However, the key difficulty in solving this problem resides in the seamless integration of the periodically updated parameters into the pre-commitment approach. This integration task is far from straightforward, as it demands reconciling the dynamic evolution of the parameters with the inherent structure of the pre-commitment framework, which was originally designed with static parameters in focus. Successfully accomplishing this integration would not only augment the realism and adaptability of the model but also offer investors more sophisticated and context-sensitive strategies, thereby filling a notable gap in the existing literature on MMV optimization.

With the aforementioned market setting and wealth process, we consider an investor who aims at maximizing the weighted combination of mean and variance of terminal wealth x_T . For each intermediate time $t \in \{0, \dots, T-1\}$, we can formulate the MMV portfolio optimization problem with time-varying risk aversion (abbreviated as TVRA-MMV) as follows,

$$\begin{aligned} \mathcal{P}(\omega_t) : \quad & \max_{\{\mathbf{u}_k\}_{k=t}^{T-1}} \mathbb{E}_t[x_T] - \omega_t \text{Var}_t[x_T] \\ & \text{s.t. } \{\mathbf{u}_t, x_t\} \text{ satisfies (4) or (5),} \end{aligned}$$

where the expected risk aversion ω_t is defined in (6). In the following part, we consider two types of market settings. In the first scenario, the capital market consists of only risky assets, then problem $\mathcal{P}(\omega_t)$ becomes the following formulation:

$$\begin{aligned} \mathcal{P}_{all-risky}(\omega_t) : \quad & \max_{\{\mathbf{u}_k\}_{k=t}^{T-1}} \mathbb{E}_t[x_T] - \omega_t \text{Var}_t[x_T] \\ & \text{s.t. } x_{k+1} = \mathbf{R}_{k+1}^\top \mathbf{u}_k, \\ & \quad x_k = \mathbf{1}^\top \mathbf{u}_k, \quad k = t, \dots, T-1. \end{aligned}$$

In the second case, there is a risk-free asset besides the risky assets, and the problem $\mathcal{P}(\omega_t)$ becomes:

$$\begin{aligned} \mathcal{P}_{risk-free}(\omega_t) : \quad & \max_{\{\mathbf{u}_k\}_{k=t}^{T-1}} \mathbb{E}_t[x_T] - \omega_t \text{Var}_t[x_T] \\ & \text{s.t. } x_{k+1} = r_{f,k} x_k + (\mathbf{R}_{k+1}^e)^\top \mathbf{u}_k, \quad k = t, \dots, T-1. \end{aligned} \tag{7}$$

Under the regime-switching model, as we have described before, we additionally define the second order moments of assets' returns $\mathbf{R}_{t+1}$ and excess returns $\mathbf{R}_{t+1}^e$ in state i as follows

$$\begin{aligned} \mathbf{Z}_t(i) &\triangleq \mathbb{E}_t[\mathbf{R}_{t+1}^e (\mathbf{R}_{t+1}^e)^\top | s_t = i] = \boldsymbol{\Sigma}_t(i) + \mathbf{r}_t^e(i) \mathbf{r}_t^e(i)^\top, \\ \mathbf{D}_t(i) &\triangleq \mathbb{E}_t[\mathbf{R}_{t+1} \mathbf{R}_{t+1}^\top | s_t = i] = \boldsymbol{\Sigma}_t(i) + \mathbf{r}_t(i) \mathbf{r}_t(i)^\top, \end{aligned} \tag{8}$$

for all $i \in \mathbb{M}$ and $t = 0, \dots, T - 1$. We will provide optimal solutions for these two problems separately in the next section.

4 Optimal Investment Strategies

This section is organized as follows. We begin in Section 4.1 by outlining the overall framework for implementing optimal strategies, encompassing both parameter estimation and strategy optimization. Section 4.2 introduces our algorithm for parameter estimation. We then derive optimal strategies for the $\mathcal{P}(\omega_t)$ problems by analyzing two distinct scenarios: portfolios consisting of only risky assets (Section 4.3.1) and portfolios including both risky and risk-free assets (Section 4.3.2). Finally, Section 4.4 explores how our strategies relate to and extend existing results in the literature.

4.1 Online implementation of portfolio policy

Existing MMV portfolio selection problems often assume the true parameters of the market, e.g., transition matrix, expected asset returns, are known in advance [3, 6, 25, 39]. However, in reality, we usually just have some prior information at first, but can update the information as new data comes in. Therefore, in this situation, estimating the parameters and implementing the strategy in an online manner seem more reasonable [45–48]. Note that our algorithm is different from existing online portfolio selection problems in that they pay more attention to predicting future asset returns and solving static optimization problem [49–51], while our paper focuses mainly on solving the MMV problem.

The process can be described as follows: at $t = 0$, we first estimate necessary parameters using historical data. The optimal policy $\mathbf{u}_0^*$ is then obtained by solving the $\mathcal{P}(\omega_0)$ problem. At subsequent time $t > 0$, as new data becomes available, parameter estimates will be updated, followed by solving the $\mathcal{P}(\omega_t)$ optimization problem again with the updated parameters. The new optimal policy $\mathbf{u}_t^*$ is then implemented. This process is iterated until $t = T - 1$. This approach ensures that parameter estimates incorporate the most recent information, allowing the policy to adapt to evolving market conditions. It is important to note that while this online process updates parameters, it does not alter the fundamental problem formulation. The whole procedure is summarized in Algorithm 1. In subsequent sections, we will delve into the methods of estimating parameters and solving the TVRA-MMV optimization problem, respectively.

Algorithm 1 Online Implementation of TVRA-MMV Policy

Require: Historical data before time $t = 0$, investment horizon T , initial wealth x_0

Ensure: Optimal portfolio allocations $\{\mathbf{u}_t^*\}_{t=0}^{T-1}$

- 1: **for** $t = 0$ to $T - 1$ **do**
 - 2: Estimate the necessary parameters based on available information up to time t .
 - 3: Solve the TVRA-MMV optimization problem $\mathcal{P}(\omega_t)$.
 - 4: Implement $\mathbf{u}_t^*$ and receive x_{t+1} .
 - 5: **end for**
-

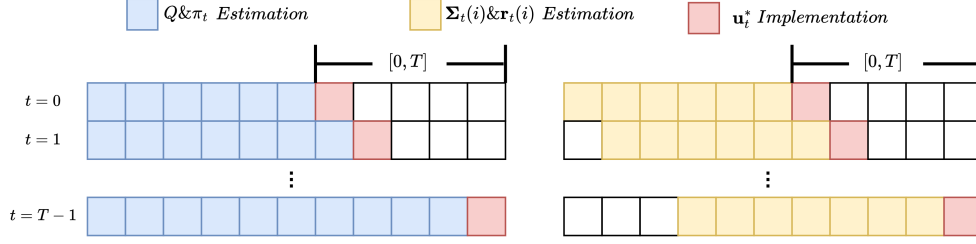


Fig. 1: Illustration of the online implementation process. The blue grid represents the expanding window for estimating the HMM model, which grows as new information becomes available. The yellow grid represents the rolling window for estimating asset return parameters, maintaining a fixed length. The red grid indicates decision-making time points.

4.2 Parameter estimation

As for the regime-switching model in this paper, assume that we do not know the true parameters of it, but initially have sufficient data before the investment horizon to train a reliable hidden Markov model. Prior to solving the TVRA-MMV problem, we should first estimate the parameters of our HMM model, which can be refined as new data becomes available. Following the methodology of Costa and Kwon [37], we employ a rolling window approach for estimating asset return parameters $\{r_{f,t}(i), \mathbf{r}_t^e(i), \mathbf{r}_t(i), \mathbf{D}_t(i), \mathbf{Z}_t(i)\}$ for $i \in \mathbb{M}$, and an expanding window approach for estimating regime-switching model parameters, such as $\mathbf{Q}$. This process is illustrated in Figure 1. The distinction between these two methods lies in their data utilization: the rolling window prioritizes recent data, providing timely market information but potentially overlooking historical trends, while the expanding window incorporates all available historical data, ensuring continuity and capturing long-term market dynamics [37]. This approach mitigates the risk of misclassifying market states due to discarding older data, as it smooths probability estimates over time.

Let $\mathbb{V}_t = (V_0, V_1, \dots, V_t)$, a subset of stochastic process $\{V_t\}_{t=0}^T$, denote a time series of accumulatively univariate observations up to time t . Our objective is to find the parameters of HMM, such as the transition matrix, that maximize the probability of observing the sequence $\mathbb{V}_t$. The parameters can typically be estimated by the Maximum Likelihood method. However, there is no analytical solution to this problem, necessitating an iterative approach starting with some initial parameter estimates. The Baum-Welch algorithm [52], a special case of the expectation-maximization (EM) algorithm, is a popular and effective approach to solving this problem. Specifically, let $\Lambda = (\mathbf{Q}, \mathbf{B}, \pi_0)$ be a tuple of parameters to be estimated, where $\mathbf{B}$ is the observation probability distribution and π_0 is the initial state distribution in the historical data. The objective is to find the optimal parameter $\Lambda^*(\mathbb{V}_t)$:

$$\Lambda^*(\mathbb{V}_t) = \arg \max_{\Lambda} \mathbb{P}(\mathbb{V}_t | \Lambda). \quad (9)$$

To emphasize the dependency on information set $\mathbb{V}_t$, the resulting estimates are denoted as $\mathbf{Q}_t, \mathbf{B}_t$ and $\pi_{0|t}$, respectively.

A comprehensive approach for hidden Markov model parameter estimation, state inference, and state-dependent return parameter calculation is presented in Algorithm 2. The Baum-Welch algorithm is employed in Phase 1 to estimate the HMM parameters (for a comprehensive survey, see [53]). Subsequently, in Phase 2, we utilize the maximum a posteriori estimation technique to infer the most probable state sequence corresponding to the observation sequence $\mathbb{V}_t$. This inference facilitates the estimation of state-dependent mean and covariance matrices from historical asset return data in Phase 3. It is noteworthy that this algorithm incorporates a rolling window for estimating state-dependent parameters ($\{r_{f,t}(i), \mathbf{r}_t^e(i), r_t(i), \mathbf{D}_t(i), \mathbf{Z}_t(i)\}$ for $i \in \mathbb{M}$). This window is typically shorter than the full observation horizon, allowing for adaptability and time-varying parameter estimates. Equipped with these parameter estimates, we are ready to compute the optimal strategies for implementation.

4.3 Optimal MMV portfolio policies

Following Algorithm 1, we solve the TVRA-MMV optimization problem at each time step t using the most recently updated parameter estimates. This section demonstrates how to obtain the optimal allocation $\mathbf{u}_t^*$ by solving $\mathcal{P}(\omega_t)$, where we are constrained to use only information available up to time t .

Since original problem $\mathcal{P}_{all-risky}(\omega_t)$ and $\mathcal{P}_{risk-free}(\omega_t)$ are non-separable in the sense of dynamic programming, we first utilize employ embedding techniques, proposed by Li and Ng [3], to construct auxiliary problems corresponding to $\mathcal{P}_{all-risky}(\omega_t)$ and $\mathcal{P}_{risk-free}(\omega_t)$, respectively. By leveraging dynamic programming techniques, we derive the value functions and optimal solutions of the auxiliary problems, which in turn facilitates the determination of the optimal solutions to the original problems. In this subsection, we will discuss the solutions to these two problems separately.

4.3.1 Portfolio with only risky assets

Following the approach in Li and Ng [3], we first introduce an auxiliary problem $\mathcal{A}_{all-risky}(\epsilon_t, \omega_t)$ corresponding to original problem $\mathcal{P}_{all-risky}(\omega_t)$, where $\epsilon_t > 0$ is an auxiliary variable. The auxiliary problem $\mathcal{A}_{all-risky}(\epsilon_t, \omega_t)$ can be formulated as a linear-quadratic stochastic optimal control problem:

$$\begin{aligned} \mathcal{A}_{all-risky}(\epsilon_t, \omega_t) : \quad & \max_{\{\mathbf{u}_t\}_{k=t}^{T-1}} \mathbb{E}_t [-\omega_t x_T^2 + \epsilon_t x_T] \\ & \text{s.t. } x_{k+1} = \mathbf{R}_{k+1}^\top \mathbf{u}_k, \\ & x_k = \mathbf{1}^\top \mathbf{u}_k, \quad k = t, t+1, \dots, T-1, \end{aligned} \tag{10}$$

where ϵ_t and ω_t are positive parameters. Unlike the original problem, this formulation eliminates the variance term, making it amenable to solution via dynamic programming due to its separable structure.

Let $v_t(x_t)$ denote the optimal value function of the truncated problem $\mathcal{A}_{all-risky}(\epsilon_t, \omega_t)$ at time t with wealth level x_t . The value function $v_k(x_k)$ must satisfy

Algorithm 2 Hidden Markov Model Parameter Estimation and State Inference

Require: Historical observations $\{V_\tau\}_{\tau=0}^t$, asset returns $\{\mathbf{R}_\tau\}_{\tau=1}^t$ or excess asset returns $\{\mathbf{R}_\tau^e\}_{\tau=1}^t$, rolling window size $rows$

Ensure: Estimated state sequence $\{s_\tau\}_{\tau=0}^t$, transition matrix $\mathbf{Q}_t$, state-dependent return parameters $(\mathbf{r}_\tau(i), \mathbf{D}_\tau(i))$ or $(\mathbf{r}_\tau^e(i), \mathbf{Z}_\tau(i))$ for $i \in \mathbb{M}$

- 1: **Phase 1: HMM Parameter Estimation**
- 2: Initialize $\Lambda = (\mathbf{Q}, \mathbf{B}, \pi)$
- 3: **repeat**
- 4: **Step 1: Compute probabilities:**
- 5: Compute forward probabilities: $\alpha_\tau(i) = P(V_0, \dots, V_\tau, s_\tau = i | \Lambda)$
- 6: Compute backward probabilities: $\beta_\tau(i) = P(V_{\tau+1}, \dots, V_t | s_\tau = i, \Lambda)$
- 7: Calculate state probabilities: $\gamma_\tau(i) = P(s_\tau = i | V_0, \dots, V_t, \Lambda) = \frac{\alpha_\tau(i)\beta_\tau(i)}{P(V_0, \dots, V_t | \Lambda)}$
- 8: Calculate transition probabilities: $\xi_\tau(i, j) = P(s_\tau = i, s_{\tau+1} = j | V_0, \dots, V_t, \Lambda) = \frac{\alpha_\tau(i)Q_t(i, j)B_j(V_{t+1})\beta_{\tau+1}(j)}{P(V_0, \dots, V_t | \Lambda)}$, where $B_j(V_{t+1}) = P(V_{t+1} | s_{t+1} = j)$ is the emission probability.
- 9: **Step 2: Update model parameters Λ :**
- 10: Update initial state distribution: $\pi_i \leftarrow \gamma_0(i)$
- 11: Update transition matrix: $Q_t(i, j) \leftarrow \frac{\sum_{\tau=0}^{t-1} \xi_\tau(i, j)}{\sum_{\tau=0}^{t-1} \gamma_\tau(i)}$
- 12: Update emission parameters (assuming Gaussian distribution):
- 13: $\mu_i \leftarrow \frac{\sum_{\tau=0}^t \gamma_\tau(i)V_\tau}{\sum_{\tau=0}^t \gamma_\tau(i)}$, $\sigma_i^2 \leftarrow \frac{\sum_{\tau=0}^t \gamma_\tau(i)(V_\tau - \mu_i)^2}{\sum_{\tau=0}^t \gamma_\tau(i)}$
- 14: **until** convergence criterion met
- 15: **Phase 2: State Inference**
- 16: **for** $\tau = 0$ to t **do**
- 17: $s_\tau \leftarrow \arg \max_i \gamma_\tau(i)$
- 18: **end for**
- 19: **Phase 3: State-dependent Return Parameter Estimation**
- 20: **for** each state $i \in \mathbb{M}$ **do**
- 21: **if** market includes risk-free asset **then**
- 22: $\mathbf{r}_\tau^e(i) \leftarrow \mathbb{E}[\mathbf{R}_{\tau+1}^e | s_\tau = i, t - rows \leq \tau \leq t]$
- 23: $\mathbf{\Sigma}_\tau(i) \leftarrow \text{Cov}[\mathbf{R}_{\tau+1}^e | s_\tau = i, t - rows \leq \tau \leq t]$
- 24: $\mathbf{Z}_\tau(i) \leftarrow \mathbf{\Sigma}_\tau(i) + \mathbf{r}_\tau^e(i)(\mathbf{r}_\tau^e(i))^\top$
- 25: **else**
- 26: $\mathbf{r}_\tau(i) \leftarrow \mathbb{E}[\mathbf{R}_{\tau+1} | s_\tau = i, t - rows \leq \tau \leq t]$
- 27: $\mathbf{\Sigma}_\tau(i) \leftarrow \text{Cov}[\mathbf{R}_{\tau+1} | s_\tau = i, t - rows \leq \tau \leq t]$
- 28: $\mathbf{D}_\tau(i) \leftarrow \mathbf{\Sigma}_\tau(i) + \mathbf{r}_\tau(i)(\mathbf{r}_\tau(i))^\top$
- 29: **end if**
- 30: **end for**

the Bellman recursive equation [54]:

$$v_k(x_k) = \max_{\mathbf{u}_k} \mathbb{E}_k[v_{k+1}(x_{k+1})] = \max_{\mathbf{u}_k} \mathbb{E}_k[v_{k+1}(\mathbf{R}_{k+1}^\top \mathbf{u}_k)], \quad (11)$$

where $k = t, \dots, T-1$. The terminal condition is given by: $v_T(x_T) = -\omega_T x_T^2 + \epsilon_T x_T$.

Through backward induction, we can solve the value function, which enables us to obtain the optimal solution to the auxiliary problem. The complete solution is presented in Theorem 1.

Theorem 1. *For a time-varying risk aversion parameter ω_t computed using (6), the optimal value function $v_t(x_t)$ of problem $\mathcal{A}_{all-risky}(\epsilon_t, \omega_t)$ is a quadratic function with respect to current wealth x_t :*

$$v_t(x_t) = -\omega_t a_t x_t^2 + \epsilon_t b_t x_t + \frac{\epsilon_t^2}{4\omega_t} c_t, \quad (12)$$

and its optimal policy at time t is given by:

$$\mathbf{u}_t^*(x_t, \epsilon_t) = \frac{\epsilon_t}{2\omega_t} \bar{\mathbf{D}}_t^{-1} (\bar{\mathbf{r}}_t - b_t \mathbf{1}) + a_t \bar{\mathbf{D}}_t^{-1} \mathbf{1} x_t, \quad (13)$$

where

$$a_t = \frac{1}{\mathbf{1}^\top \bar{\mathbf{D}}_t^{-1} \mathbf{1}}, \quad b_t = \frac{\mathbf{1}^\top \bar{\mathbf{D}}_t^{-1} \bar{\mathbf{r}}_t}{\mathbf{1}^\top \bar{\mathbf{D}}_t^{-1} \mathbf{1}}, \quad c_t = \bar{\alpha}_t + \bar{\mathbf{r}}_t^\top \bar{\mathbf{D}}_t^{-1} \bar{\mathbf{r}}_t - \frac{(\mathbf{1}^\top \bar{\mathbf{D}}_t^{-1} \bar{\mathbf{r}}_t)^2}{\mathbf{1}^\top \bar{\mathbf{D}}_t^{-1} \mathbf{1}}, \quad (14)$$

and the parameters $\bar{\mathbf{r}}_t, \bar{\mathbf{D}}_t, \bar{\alpha}_t$ can be computed by running Algorithm 3 based on $\mathcal{F}_t$.

Remark 1. *The parameters $\bar{\mathbf{r}}_t, \bar{\mathbf{D}}_t$, and $\bar{\alpha}_t$ can be roughly interpreted as probability-weighted averages of their corresponding state-dependent estimates based on all available information up to t . In this case, the weighting factor is the probability of the system being in state i at time t , for all $i \in \mathbb{M}$. This probabilistic interpretation provides insight into the structure of these parameters and their role in capturing the dynamics of the regime-switching market model.*

We provide a useful lemma to illustrate the properties of some parameters in Algorithm 3, which helps to prove Theorem 1.

Lemma 1. *For all market states $i \in \mathbb{M}$ and time periods $t \in \{0, \dots, T-1\}$, the following properties hold: i) The state-dependent parameters satisfy $\tilde{\gamma}_t(i) > 0$ and $\hat{\gamma}_t(i) > 0$; ii) The matrix $\bar{\mathbf{D}}_t$ is positive definite; iii) The parameter a_t is strictly positive.*

The positive definiteness of $\bar{\mathbf{D}}_t$ guarantees that the covariance structure of asset returns remains non-degenerate across all market regimes, a common condition for meaningful portfolio optimization. Moreover, the positivity of a_t and $\bar{\mathbf{D}}_t$ ensures that the quadratic term in the value function $v_t(x_t)$ remains concave, which is essential for the existence of a unique optimal solution. With the help of parameters calculated in Algorithm 3, we can derive the value function and optimal solution of $\mathcal{A}_{all-risky}(\epsilon_t, \omega_t)$, as is stated in Theorem 1. The proofs of Theorem 1 and Lemma 1 can be found in Appendix A.

Having the optimal policy of auxiliary problem $\mathcal{A}_{all-risky}(\epsilon_t, \omega_t)$, we can derive the first and second order moments with respect to the terminal wealth, which will be stated in the following theorem.

Theorem 2. *When implementing the optimal strategy (13) for the auxiliary problem $\mathcal{A}_{all-risky}(\epsilon_t, \omega_t)$, the first and second order moments of the terminal wealth x_T are*

Algorithm 3 Parameter Calculation for Auxiliary Problem $\mathcal{A}_{all-risky}(\epsilon_t, \omega_t)$ and Original Problem $\mathcal{P}_{all-risky}(\omega_t)$

Require: transition matrix $\mathbf{Q}_t$, state distribution π_t at time t and state-dependent return parameters $\mathbf{r}_k(i), \mathbf{D}_k(i)$ for $i \in \mathbb{M}$ estimated based on $\mathcal{F}_t$

Ensure: Parameters $\{\bar{\mathbf{r}}_t, \bar{\mathbf{D}}_t, \bar{\alpha}_t\}$

```

1: Initialize  $\hat{\gamma}_T(i) = \hat{\rho}_T(i) = \hat{\gamma}_T(i) = \hat{\rho}_T(i) = 1, \bar{\alpha}_T(i) = \hat{\alpha}_T(i) = 0$  for all  $i \in \mathbb{M}$ ,
    $d_T = j_T = 1$ 
2: for  $k = T - 1, \dots, t$  do
3:   for each  $i \in \mathbb{M}$  do
4:      $\hat{\gamma}_{k+1}(i) \leftarrow \sum_{j \in \mathbb{M}} Q_t(i, j) \hat{\gamma}_{k+1}(j)$ 
5:      $\hat{\rho}_{k+1}(i) \leftarrow \sum_{j \in \mathbb{M}} Q_t(i, j) \hat{\rho}_{k+1}(j)$ 
6:      $\hat{\alpha}_{k+1}(i) \leftarrow \sum_{j \in \mathbb{M}} Q_t(i, j) \hat{\alpha}_{k+1}(j)$ 
7:   end for
8:   Compute  $\pi_k = \pi_t \mathbf{Q}_t^{k-t}$ .
9:    $\bar{\mathbf{r}}_k \leftarrow \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \hat{\rho}_{k+1}(i) \mathbf{r}_k(i)$ 
10:   $\bar{\mathbf{D}}_k \leftarrow \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \hat{\gamma}_{k+1}(i) \mathbf{D}_k(i)$ 
11:   $\bar{\alpha}_k \leftarrow \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \hat{\alpha}_{k+1}(i)$ 
12:   $d_k \leftarrow \frac{1}{\mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{1}}$ 
13:   $j_k \leftarrow \frac{\mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \bar{\mathbf{r}}_k}{\mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{1}}$ 
14:  for each  $i \in \mathbb{M}$  do
15:     $\hat{\gamma}_k(i) \leftarrow \hat{\gamma}_{k+1}(i) d_k^2 \mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{D}_k(i) \bar{\mathbf{D}}_k^{-1} \mathbf{1}$ 
16:     $\hat{\rho}_k(i) \leftarrow \hat{\gamma}_{k+1}(i) d_k j_k \mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{D}_k(i) \bar{\mathbf{D}}_k^{-1} \mathbf{1}$ 
17:     $\bar{\alpha}_k(i) \leftarrow -\hat{\gamma}_{k+1}(i) j_k^2 \mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{D}_k(i) \bar{\mathbf{D}}_k^{-1} \mathbf{1} - \hat{\gamma}_{k+1}(i) \bar{\mathbf{r}}_k^\top \bar{\mathbf{D}}_k^{-1} \mathbf{D}_k(i) \bar{\mathbf{D}}_k^{-1} \bar{\mathbf{r}}_k$ 
18:     $+ 2\hat{\gamma}_{k+1}(i) j_k \mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{D}_k(i) \bar{\mathbf{D}}_k^{-1} \bar{\mathbf{r}}_k + 2\hat{\rho}_{k+1}(i) \bar{\mathbf{r}}_k^\top \bar{\mathbf{D}}_k^{-1} \mathbf{r}_k(i)$ 
19:     $- 2\hat{\rho}_{k+1}(i) j_k \mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \bar{\mathbf{r}}_k \mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{r}_k(i) + \hat{\alpha}_{k+1}(i)$ 
20:  end for
21: end for

```

given by:

$$\mathbb{E}_t[x_T] = b_t x_t + \frac{\epsilon_t}{2\omega_t} c_t, \quad \mathbb{E}_t[x_T^2] = a_t x_t^2 + \frac{\epsilon_t^2}{4\omega_t^2} c_t, \quad (15)$$

where a_t, b_t , and c_t are parameters defined in equation (14).

The proof is presented in Appendix A. To complete the solution, we still need to find the optimal value of the auxiliary variable ϵ_t in policy (13). By determining ϵ_t^* , we can derive the optimal strategy for solving the original problem $\mathcal{P}_{all-risky}(\omega_t)$. By applying Theorem 2, when implementing the optimal strategy for the auxiliary problem $\mathcal{A}_{all-risky}(\epsilon_t, \omega_t)$, the variance of the terminal wealth x_T can be derived as follows:

$$\text{Var}_t[x_T] = (a_t - b_t^2) x_t^2 - \epsilon_t \frac{b_t}{\omega_t} c_t x_t + \frac{\epsilon_t^2}{4\omega_t^2} (1 - c_t) c_t. \quad (16)$$

To ensure the model remains meaningful when current wealth x_t is zero, the terminal wealth variance must be non-zero. This leads to the following lemma:

Lemma 2. *The parameter c_t computed in Theorem 1 satisfies the inequality $0 < c_t < 1$.*

Therefore, at time t the utility function of problem $\mathcal{P}_{all-risky}(\omega_t)$ can be written as a quadratic function with respect to parameter ϵ_t

$$\begin{aligned} U_t(\epsilon_t) &= \mathbb{E}_t[x_T] - \omega_t \text{Var}_t[x_T] \\ &= b_t x_t + \frac{\epsilon_t}{2\omega_t} c_t - \omega_t (a_t - b_t^2) x_t^2 + \epsilon_t b_t c_t x_t - \frac{\epsilon_t^2}{4\omega_t} (1 - c_t) c_t. \end{aligned} \quad (17)$$

By using Lemma 2, we can verify the concavity of the utility function since

$$U_t''(\epsilon_t) = -\frac{(1 - c_t)c_t}{2\omega_t} < 0. \quad (18)$$

That is to say, a unique maximizer of $U_t(\epsilon_t)$ exists, and it should satisfy the first-order optimal condition, i.e.,

$$U_t'(\epsilon_t) = \frac{1}{2\omega_t} c_t + b_t c_t x_t - \frac{2\epsilon_t}{4\omega_t} (1 - c_t) c_t = 0. \quad (19)$$

Solving this equation gives the optimal parameter value

$$\epsilon_t^* = \frac{1 + 2\omega_t b_t x_t}{1 - c_t}. \quad (20)$$

By substituting the optimal ϵ_t^* back into the auxiliary problem solution (13), the optimal solution to the original problem $\mathcal{P}_{all-risky}(\omega_t)$ is straightforward and the following theorem formally states the result.

Theorem 3. *For any time period $t \in \{0, \dots, T - 1\}$, given current wealth x_t , the optimal wealth allocation $\mathbf{u}_t^*$ is:*

$$\mathbf{u}_t^*(x_t) = \left(\frac{1 + 2\omega_t b_t x_t}{2(1 - c_t)\omega_t} \right) \bar{\mathbf{D}}_t^{-1} (\bar{\mathbf{r}}_t - b_t \mathbf{1}) + a_t \bar{\mathbf{D}}_t^{-1} \mathbf{1} x_t, \quad (21)$$

where ω_t is the time-varying risk aversion parameter, a_t, b_t, c_t are parameters defined in equation (14) and $\bar{\mathbf{D}}_t, \bar{\mathbf{r}}_t$ are computed recursively using Algorithm 3.

Note that within the online framework, once new information arrives, the estimates of $\mathbf{Q}$ and state probability will be updated, thus the parameters, as well as the policy, should be readjusted.

4.3.2 Portfolio with both risky and risk-free assets

Adopting a similar approach as the previous case, in the market where a risk-free asset is involved, for each stage $t \in \{0, \dots, T - 1\}$, we first formulate an auxiliary problem

$\mathcal{A}_{risk-free}(\lambda_t, \omega_t)$ corresponding to the problem $\mathcal{P}_{risk-free}(\omega_t)$,

$$\begin{aligned} \mathcal{A}_{risk-free}(\lambda_t, \omega_t) : \quad & \max_{\{\mathbf{u}_k\}_{k=t}^{T-1}} \mathbb{E}_t [-\omega_t x_T^2 + \lambda_t x_T] \\ \text{s.t. } & x_{k+1} = r_{f,k} x_k + (\mathbf{R}_{k+1}^e)^\top \mathbf{u}_k, \quad k = t, \dots, T-1. \end{aligned} \quad (22)$$

Define $\kappa_t(x_t)$ as the optimal value function of truncated problem $\mathcal{A}_{risk-free}(\lambda_t, \omega_t)$, then $\kappa_t(x_t)$ should satisfies the recursive equation as follows,

$$\kappa_k(x_k) = \max_{\mathbf{u}_k} \mathbb{E}_k [\kappa_{k+1}(x_{k+1})] = \max_{\mathbf{u}_k} \mathbb{E}_k [\kappa_{k+1}(r_{f,k} x_k + (\mathbf{R}_{k+1}^e)^\top \mathbf{u}_k)], \quad (23)$$

for $k = t, \dots, T-1$, with the terminal condition $\kappa_T(x_T) = -\omega_t x_T^2 + \lambda_t x_T$.

As we will present in Theorem 4 and Algorithm 4, the value function can be solved by backward induction, providing an explicit solution to the auxiliary problem $\mathcal{A}_{risk-free}(\lambda_t, \omega_t)$.

Theorem 4. *For a time-varying risk aversion parameter ω_t computed using (6), the optimal value function $\kappa_t(x_t)$ of problem $\mathcal{A}_{risk-free}(\lambda_t, \omega_t)$ is a quadratic function with respect to current wealth x_t :*

$$\kappa_t(x_t) = -\omega_t h_t x_t^2 + \lambda_t p_t x_t + \frac{\lambda_t^2}{4\omega_t} q_t, \quad (24)$$

and its optimal policy at time t is given by:

$$\mathbf{u}_t^*(x_t, \lambda_t) = \frac{\lambda_t}{2\omega_t} \bar{\mathbf{Z}}_t^{-1} \bar{\mathbf{r}}_{e,t} - \bar{\mathbf{Z}}_t^{-1} \bar{\mathbf{r}}_{ef,t} x_t, \quad (25)$$

where

$$h_t = \bar{r}_{2f,t} - \bar{\mathbf{r}}_{ef,t}^\top \bar{\mathbf{Z}}_t^{-1} \bar{\mathbf{r}}_{ef,t}, \quad p_t = \bar{r}_{f,t} - \bar{\mathbf{r}}_{ef,t}^\top \bar{\mathbf{Z}}_t^{-1} \bar{\mathbf{r}}_{e,t}, \quad q_t = \bar{\eta}_t + \bar{\mathbf{r}}_{e,t}^\top \bar{\mathbf{Z}}_t^{-1} \bar{\mathbf{r}}_{e,t}, \quad (26)$$

and the parameters $\bar{r}_{2f,t}, \bar{\mathbf{r}}_{ef,t}, \bar{\mathbf{r}}_{e,t}, \bar{r}_{f,t}, \bar{\mathbf{Z}}_t, \bar{\eta}_t$ can be computed by running Algorithm 4 based on $\mathcal{F}_t$.

The proof of Theorem 4 follows a similar approach to that of Theorem 1, with the key difference being the wealth evolution process. Furthermore, we present Lemma 3 which can be used to prove Theorem 4 and other related results. The detailed proofs of Theorem 4 and Lemma 3 can be found in Appendix B.

Lemma 3. *For all market states $i \in \mathbb{M}$ and time periods $t \in \{0, \dots, T-1\}$, the following properties hold: i) The state-dependent parameters satisfy $\tilde{\omega}_t(i) > 0$ and $\hat{\omega}_t(i) > 0$; ii) The matrix $\bar{\mathbf{Z}}_t$ is positive definite; iii) The parameter h_t is strictly positive.*

Similar to the property of $\bar{\mathbf{D}}_t$ and a_t , the positive definiteness of $\bar{\mathbf{Z}}_t$ and the positivity of h_t guarantees that the existence of a unique optimal solution of the value function $\kappa_t(x_t)$. We proceed to derive the optimal strategy for $\mathcal{P}_{risk-free}(\omega_t)$ by determining the optimal value for parameter λ_t , with the help of relationship between the optimal strategy for $\mathcal{A}_{risk-free}(\lambda_t, \omega_t)$ and that of $\mathcal{P}_{risk-free}(\omega_t)$.

Algorithm 4 Computation of Parameters for Auxiliary Problem $\mathcal{A}_{risk-free}(\lambda_t, \omega_t)$ and Original Problem $\mathcal{P}_{risk-free}(\omega_t)$

Require: transition matrix $\mathbf{Q}_t$, state distribution π_t at time t and state-dependent return parameters $r_{f,k}(i), \mathbf{r}_k^e(i), \mathbf{Z}_k(i)$ for $i \in \mathbb{M}$ estimated based on all available information up to time t .

Ensure: Parameters $\bar{r}_{2f,t}, \bar{\mathbf{r}}_{ef,t}, \bar{\mathbf{r}}_{e,t}, \bar{r}_{f,t}, \bar{\mathbf{Z}}_t, \bar{\eta}_t$

```

1: Initialize  $\tilde{\omega}_T(i) = \tilde{\lambda}_T(i) = \tilde{\omega}_T(i) = \hat{\lambda}_T(i) = 1, \tilde{\eta}_T(i) = 0, \hat{\eta}_T(i) = 0$  for all  $i \in \mathbb{M}$ 
2: for  $k = T-1, T-2, \dots, t$  do
3:   for each  $i \in \mathbb{M}$  do
4:      $\hat{\omega}_{k+1}(i) \leftarrow \sum_{j \in \mathbb{M}} Q_t(i, j) \tilde{\omega}_{k+1}(j)$ 
5:      $\hat{\lambda}_{k+1}(i) \leftarrow \sum_{j \in \mathbb{M}} Q_t(i, j) \tilde{\lambda}_{k+1}(j)$ 
6:      $\hat{\eta}_{k+1}(i) \leftarrow \sum_{j \in \mathbb{M}} Q_t(i, j) \tilde{\eta}_{k+1}(j)$ 
7:   end for
8:   Compute  $\pi_k = \pi_t \mathbf{Q}_t^{k-t}$ .
9:    $\bar{r}_{2f,k} \leftarrow \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \hat{\omega}_{k+1}(i) r_{f,k}(i)^2$ 
10:   $\bar{\mathbf{r}}_{ef,k} \leftarrow \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \hat{\omega}_{k+1}(i) r_{f,k}(i) \mathbf{r}_k^e(i)$ 
11:   $\bar{\mathbf{r}}_{e,k} \leftarrow \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \hat{\lambda}_{k+1}(i) \mathbf{r}_k^e(i)$ 
12:   $\bar{r}_{f,k} \leftarrow \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \hat{\lambda}_{k+1}(i) r_{f,k}(i)$ 
13:   $\bar{\mathbf{Z}}_k \leftarrow \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \hat{\omega}_{k+1}(i) \mathbf{Z}_k(i)$ 
14:   $\bar{\eta}_k \leftarrow \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \hat{\eta}_{k+1}(i)$ 
15:  for each  $i \in \mathbb{M}$  do
16:     $\tilde{\omega}_k(i) \leftarrow \hat{\omega}_{k+1}(i) r_{f,k}(i)^2 - 2 \hat{\omega}_{k+1}(i) r_{f,k}(i) \mathbf{r}_k^e(i)^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k}$ 
17:     $\tilde{\lambda}_k(i) \leftarrow \hat{\lambda}_{k+1}(i) r_{f,k}(i) - \hat{\lambda}_{k+1}(i) \mathbf{r}_k^e(i)^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} - \hat{\omega}_{k+1}(i) r_{f,k}(i) \mathbf{r}_k^e(i)^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{e,k}$ 
18:     $\tilde{\eta}_k(i) \leftarrow 2 \hat{\lambda}_{k+1}(i) \mathbf{r}_k^e(i)^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{e,k} - \hat{\omega}_{k+1}(i) \bar{\mathbf{r}}_{e,k}^\top \bar{\mathbf{Z}}_k^{-1} \mathbf{Z}_k(i) \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{e,k} + \hat{\eta}_{k+1}(i)$ 
19:  end for
20: end for

```

Theorem 5. When implementing the optimal strategy (25) for the auxiliary problem $\mathcal{A}_{risk-free}(\lambda_t, \omega_t)$, the first and second order moments of the terminal wealth x_T are given by:

$$\mathbb{E}_t[x_T] = p_t x_t + \frac{\lambda_t}{2\omega_t} q_t, \quad \mathbb{E}_t[x_T^2] = h_t x_t^2 + \frac{\lambda_t^2}{4\omega_t^2} q_t, \quad (27)$$

where p_t, h_t , and q_t are parameters defined in equation (26).

By applying Theorem 5, the variance of the terminal wealth can be derived as follows:

$$\text{Var}_t[x_T] = (h_t - p_t^2) x_t^2 - \lambda_t \frac{p_t}{\omega_t} q_t x_t + \frac{\lambda_t^2}{4\omega_t^2} (1 - q_t) q_t. \quad (28)$$

To make sure the model remain meaningful when current wealth x_t is zero, the terminal wealth variance must be non-zero, which implies the following lemma:

Lemma 4. The parameter q_t computed in Algorithm 4 satisfies the inequality $0 < q_t < 1$.

The proofs of Theorem 5 and Lemma 4 are similar to those of Theorem 2 and Lemma 2, thus we omit here. The objective function of problem $(\mathcal{P}_{risk-free}(\omega_t))$ can be written as a quadratic function in terms of λ_t :

$$\begin{aligned} U_t(\lambda_t) &= \mathbb{E}_t(x_T) - \omega_t \text{Var}_t(x_T) \\ &= p_t x_t + \frac{\lambda_t}{2\omega_t} q_t - \omega_t (h_t - p_t^2) x_t^2 + \lambda_t p_t q_t x_t - \frac{\lambda_t^2}{4\omega_t} (1 - q_t) q_t. \end{aligned} \quad (29)$$

Using Lemma 4, we can confirm that this function is concave by examining its second derivative:

$$U_t''(\lambda_t) = -\frac{(1 - q_t)q_t}{2\omega_t} < 0. \quad (30)$$

Applying the first-order optimality condition to maximize $U_t(\lambda_t)$ yields a unique optimal value:

$$\lambda_t^* = \frac{1 + 2\omega_t p_t x_t}{1 - q_t}. \quad (31)$$

Substituting the optimal λ_t^* back into the auxiliary problem solution (25), the optimal solution to the original problem is straightforward, which is stated formally in the following theorem.

Theorem 6. *For any time period $t \in \{0, \dots, T - 1\}$, given current wealth x_t , the optimal portfolio allocation $\mathbf{u}_t^*$ is:*

$$\mathbf{u}_t^*(x_t) = \left(\frac{1 + 2\omega_t p_t x_t}{2(1 - q_t)\omega_t} \right) \bar{\mathbf{Z}}_t^{-1} \bar{\mathbf{r}}_{e,t} - \bar{\mathbf{Z}}_t^{-1} \bar{\mathbf{r}}_{ef,t} x_t, \quad (32)$$

where ω_t is the time-varying risk aversion parameter, p_t, q_t are parameters defined in equation (26) and $\bar{\mathbf{Z}}_t, \bar{\mathbf{r}}_{e,t}, \bar{\mathbf{r}}_{ef,t}$ are computed recursively using Algorithm 4.

While we adopt an online algorithm to update the parameters in Theorem 3 and 6, it degenerates to the classical offline solutions presented in [6, 25, 55] if new information is not revealed.

4.4 Some special settings

In Section 4.3.1 and 4.3.2, we propose a MMV optimization model constructed based on a hidden Markov model and a state-dependent risk aversion coefficient. This model is designed to be versatile, capable of accommodating a variety of special models that have been considered in existing literature. In this subsection, we discuss some special problem formulations by either relaxing or strengthening the assumptions of our model.

If the market regime is fully observable at any given period, i.e., the state s_t is known with certainty at time t , the underlying model simplifies to a Markov chain. Consequently, when conditioning on $s_t = i$, the parameters in the value function defined in Theorem 4 become state-dependent and reduce to:

$$\begin{aligned}
\tilde{\omega}_t(i) &= \hat{\omega}_{t+1}(i) r_{f,t}(i)^2 [1 - \mathbf{r}_t^e(i)^\top \mathbf{Z}_t^{-1}(i) \mathbf{r}_t^e(i)], \\
\tilde{\lambda}_t(i) &= \hat{\lambda}_{t+1}(i) r_{f,t}(i) [1 - \mathbf{r}_t^e(i)^\top \mathbf{Z}_t^{-1}(i) \mathbf{r}_t^e(i)], \\
\tilde{\eta}_t(i) &= \frac{\hat{\lambda}_{t+1}^2(i)}{\hat{\omega}_{t+1}(i)} \mathbf{r}_t^e(i)^\top \mathbf{Z}_t^{-1}(i) \mathbf{r}_t^e(i) + \hat{\eta}_{t+1}(i).
\end{aligned} \tag{33}$$

Under these conditions, the optimal parameter λ_t^* is given by:

$$\lambda_t^* = \frac{1 + 2\omega_t \tilde{\lambda}_0(i) x_0}{1 - \tilde{\eta}_0(i)}, \tag{34}$$

and the optimal strategy can be expressed as:

$$\mathbf{u}_t^*(x_t) = \left[\frac{1}{2} \left(\frac{1 + 2\omega_t \tilde{\lambda}_0(i) x_0}{(1 - \tilde{\eta}_0(i)) \omega_t} \right) \frac{\hat{\lambda}_{t+1}}{\hat{\omega}_{t+1}} - r_{f,t}(i) x_t \right] \mathbf{Z}_t^{-1}(i) \mathbf{r}_t^e(i). \tag{35}$$

Remark 2. *If the risk aversion remains constant throughout the investment horizon, i.e., $\omega_t = \omega$ irrespective of the market regime s_t , then it can be readily verified that our results align with those presented in Çakmak and Özekici [6].*

If the market is characterized by a single regime, the returns of risky assets are independent and identically distributed, and the risk aversion coefficient remains constant throughout the investment horizon, the optimal parameter λ_t^* simplifies to:

$$\lambda_t^* = 2\omega_t x_0 \prod_{t=0}^{T-1} r_{f,t} + \frac{1}{\prod_{t=0}^{T-1} (1 - (\mathbf{r}_t^e)^\top \mathbf{Z}_t^{-1} \mathbf{r}_t^e)}. \tag{36}$$

Consequently, the optimal strategy can be expressed as:

$$\mathbf{u}_t^*(x_t) = \left[\left(x_0 \prod_{t=0}^{T-1} r_{f,t} + \frac{1}{2\omega_t \prod_{t=0}^{T-1} (1 - (\mathbf{r}_t^e)^\top \mathbf{Z}_t^{-1} \mathbf{r}_t^e)} \right) \prod_{k=t+1}^{T-1} \frac{1}{r_{f,k}} - x_t r_{f,t} \right] \mathbf{Z}_t^{-1} \mathbf{r}_t^e. \tag{37}$$

These results are consistent with those derived by Li and Ng [3].

While our work assumes that the regime-switching model can be described by an HMM, the research of Gao and Li [25] examined a more generalized setting where asset returns are generally correlated. Assuming a fixed risk aversion coefficient, our model can be interpreted as a specific case within their general framework. In particular, when the correlation is characterized by a hidden Markov model, the parameters in their work can be explicitly expressed, which are equivalent to our findings. As a result, the optimal policies in our model can be derived by directly applying their solutions with our specified parameters. Consequently, our work can be viewed as an extension of Gao and Li [25], incorporating time-varying risk aversion coefficients within the context of hidden Markov models.

5 Numerical analysis

In this section, we present a series of numerical experiments to demonstrate the effectiveness of the proposed model by comparing it with several benchmark models. To start with, in Section 5.1, we utilize monthly VIX index data spanning from January 2000 to December 2023, which is also used in Li et al. [16], to determine the appropriate number of market states. We employ Gaussian HMM to estimate the number of market states and calibrate the corresponding state transition matrix. Based on the results, we estimate the monthly mean returns and covariance matrix with respect to asset returns for each market state $i \in \mathbb{M}$, and then compute parameters $\mathbf{r}_t(i)$, $\mathbf{r}_t^e(i)$, $\mathbf{Z}_t(i)$, $\mathbf{D}_t(i)$. Meanwhile, we can calculate the multi-step transition probability by (2) to predict the regime shifts in the future. Equipped with these estimates, we are able to compute the necessary parameters following Algorithm 3 and 4, respectively, which will be utilized to formulate investment strategy. In Section 5.2, we illustrate the trading characteristics of TVRA-MMV by examining the changes in risk aversion parameters and the optimal allocations of one risky asset for varying market states in one typical sample path. Finally, in Section 5.3, we conduct out-of-sample experiments, comparing various performance metrics of the optimal strategies derived from our models as well as other benchmark models. The results highlight the superior performance of our proposed models in time-varying market environments.

5.1 Calibration of regime-switching market

Identifying appropriate market regimes is crucial for accurate modeling in financial economics, particularly when using regime-switching models. A key challenge in regime-switching models is to determine the optimal number of regimes. Overestimating the number of regimes introduces unnecessary parameters, leading to increased estimation errors, while underestimating them may result in a model that inadequately captures the true dynamics of the market [37]. This subsection aims to evaluate both two-state and three-state regime-switching models to ascertain which one better describes market conditions for subsequent analysis.

Our analysis employs Gaussian HMM to examine the dynamic changes in market conditions. Specifically, we utilize the Baum–Welch algorithm [52] to estimate the probability distributions of the hidden states. We hypothesize that the market operates under either two or three distinct market regimes (also can be extended to more states), then use the VIX index data³ as observations, which is widely recognized as a barometer of market volatility and extensively employed to represent market fluctuations, to estimate the market states and state transition matrices for both models using an expanding window approach.

Figure 2(a) presents the estimation results of the two-state model. In this figure, yellow dots represent periods identified as economic recessions, while dark blue dots indicate periods of economic growth. Figure 2(b) shows the estimation results of the three-state model, additionally introducing a green state that denotes a stable period between recession and growth. Analyzing the estimation results, we observe that both models tend to classify intervals with lower VIX indices as periods of economic growth

³The VIX index data can be found from https://www.cboe.com/tradable_products/vix/.

and intervals with higher VIX indices as periods of economic recession. Notably, short-term fluctuations in the VIX index do not necessarily trigger a regime switch in the models. Instead, a significant and sustained change in the VIX level is required for the models to identify a transition in market state. This behavior aligns with economic realities, where prolonged periods of high volatility are indicative of recessions rather than brief market perturbations.

Comparing the results of these two models, the two-state model demonstrates a superior ability to capture major economic downturns, such as the 2008 global financial crisis and the 2020 COVID-19 pandemic. It effectively delineates periods of recession and subsequent recovery, reflecting the sustained impact of these events on the economy. In contrast, the three-state model, despite introducing an intermediate stable state, appears less effective in representing the prolonged effects of significant economic disruptions. The additional state may complicate the model without providing substantial explanatory power regarding market transitions.

In conclusion, the two-state regime-switching model is more suitable for describing market conditions in our analysis, due to its simplicity and effectiveness in capturing the essential features of market dynamics. This kind of two-state regime-switching model is also widely used in literature, where the two market states are commonly referred to as bull and bear markets, see [7, 9, 16]. In subsequent experiments, we will use State 0 or bear market to represent the state of economic recession, and use State 1 or bull market to represent the state of economic growth.

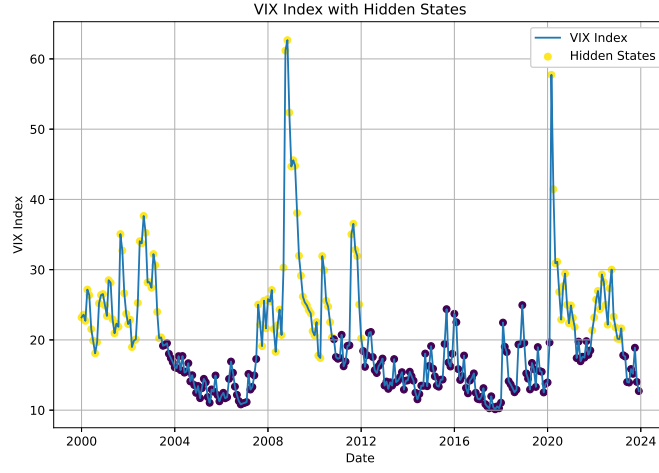
5.2 Comparison of portfolio policies

In this section, we examine optimal allocation strategies under the two distinct investment scenarios. To simplify the analysis, we assume the market consists of only two assets. In the first scenario, detailed in Section 4.3.1, both assets are risky: one asset exhibits high volatility and high expected return, while the other has low volatility and low expected return. In the second scenario, discussed in Section 4.3.2, the market comprises one risky asset and one risk-free asset. For both scenarios, we solve TVRA-MMV problems formulated as $P_e(\bar{\omega})$ and $P_f(\bar{\omega})$ by utilizing the optimal strategies derived in Theorems 3 and 6, respectively. Note that for the experiments conducted in Section 5.2 and 5.3, we select historical assets' returns data from January 2000 to December 2023 in alignment with the time span of VIX index data, and calibrate parameters (mean and covariance matrix) for each estimated market regimes. The initial wealth is set to be $x_0 = 1$ and the investment horizon has $T = 6$ periods. We generate trajectories of market states based on the state transition matrix and simulate the assets' returns according to the generate sample states using the expected returns and covariance matrices.

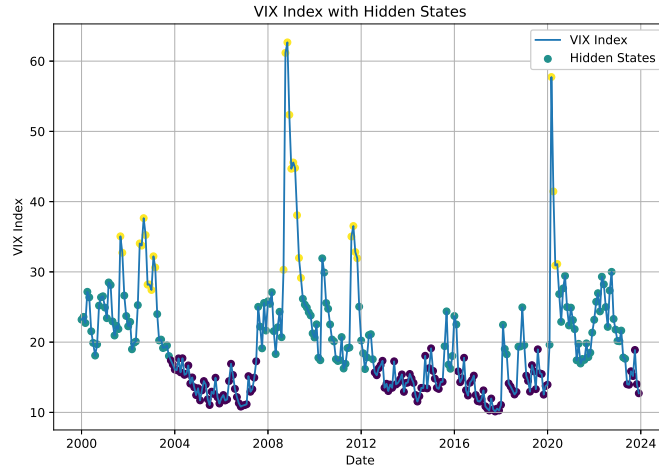
For the experiments involving two risky assets, we utilize the monthly returns of the S&P 500⁴ and the DAX indices⁵. The S&P 500 is widely regarded as a benchmark for large-cap U.S. equities, while the DAX index represents 40 of the largest and most liquid German companies traded on the Frankfurt Stock Exchange. These two

⁴Monthly observations of S&P 500 can be found from the Center for Research in Security Prices (CRSP) database.

⁵The DAX index data can be found from <https://www.boerse-frankfurt.de/indices/dax>.



(a) Two-state model.



(b) Three-state model.

Fig. 2: Comparison of the two-state and three-state regime switching models calibrated using VIX index data from 2000 to 2023. The two-state model identifies periods of recession (yellow) and growth (dark blue), while the three-state model introduces a stable state (green) between recession and growth.

datasets are extensively used in empirical research (see [16, 56]). By estimating the means and covariance matrix of these two assets' returns based on historical sample data, we observe that the DAX index exhibits higher expected return and higher volatility compared to the S&P 500. For the experiment involving one risky asset and one risk-free asset, we consider the S&P 500 as the risky asset and U.S. Treasury bills (T-bills)⁶ as the risk-free asset. T-bills are short-term debt obligations backed by the U.S. Department of the Treasury and are commonly used as proxies for the risk-free asset in the literature [16, 27].

We compare the performance of different investment strategies implemented by five types of investors, including our proposed model and four benchmark models. The detailed settings are as follows:

- **i.i.d. investors:** These investors believe that asset returns are independently and identically distributed (i.i.d.) across periods, unaware of regime-switching dynamics. They are characterized by different levels of risk aversion:
 - **aggressive investor (i.i.d. ($\omega = 2$)):** Employs a low risk aversion coefficient.
 - **conservative investor (i.i.d. ($\omega = 10$)):** Employs a high risk aversion coefficient.
 - **moderate investor (i.i.d. ($\omega = 5$)):** Employs a risk aversion that is a weighted combination of the previous two based on the stationary probability of the Markov chain, reflecting a moderate attitude toward changing market conditions.

Their respective risk aversion parameters are determined at time $t = 0$ and remain constant throughout the investment horizon.

- **TVRA-MMV(MC):** An “oracle” investor who is fully aware of the market’s regime-switching dynamics and can accurately identify the current market state at each period, effectively knowing the true Markov chain state. This investor employs regime-dependent risk aversion coefficients, setting $\omega_1 = 2$ during bull market conditions and $\omega_2 = 10$ during bear market conditions. This is an ideal setting since no one can have perfect information about the market state.
- **TVRA-MMV(HMM):** Our proposed strategy, which represents an investor who knows that the market operates under two regimes but cannot directly observe the current state. Instead, the investor models the market shifts using HMM and uses the VIX index as observations to calibrate the probabilities of being in each regime. This investor also employs regime-dependent risk aversion coefficients, with $\omega_1 = 2$ in the bull market and $\omega_2 = 10$ in the bear market.

To illustrate their performance and adaptability in varying market conditions, we choose one typical sample path that includes market regime changes as an example. Figure 3(a) depicts the market regimes within our selected investment horizon. Figure 3(b) compares the risk aversion coefficients of five different investors, from which we can see that while i.i.d. investors’ risk profile remain constant, TVRA-MMV(MC) and TVRA-MMV(HMM) change their attitude as market state

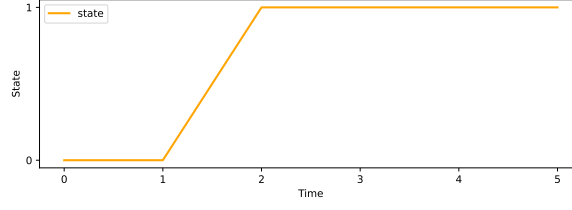
⁶The U.S. T-bill rates can be found from the Kenneth R. French website <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/>.

changes. TVRA-MMV(HMM) gradually learns the true market state and follows TVRA-MMV(MC).

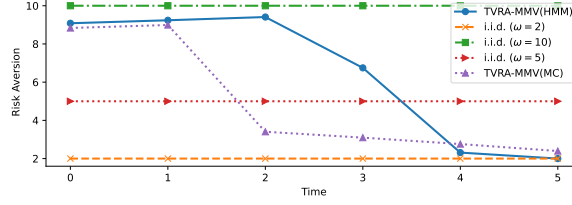
We investigate the optimal capital allocation in the risky asset S&P 500 in each period by solving these five models, as is shown in Figure 3(c) and 3(d). For investors who are not aware of the regime-switching dynamics (i.e., i.i.d. strategies with $\omega = 2$, $\omega = 5$ and $\omega = 10$), the optimal allocations remain relatively stable over time, shown as the yellow and green lines in both Figures 3(c) and 3(d). The investor with lower risk aversion (i.i.d. ($\omega = 2$)) adopts a more aggressive strategy compared to the investor with higher risk aversion (i.i.d. ($\omega = 5$) and i.i.d. ($\omega = 10$)), resulting in consistently higher allocations to the S&P 500. This outcome is consistent with classical portfolio theory, which states that lower risk aversion indicates a preference for higher expected returns regardless of the associated risks. However, none of these i.i.d. strategies responds to changes in market regimes. As illustrated in Figures 3(c) and 3(d), the allocations to the S&P 500 index remain nearly constant even when the market transitions from “bear” to “bull” states. This lack of responsiveness indicates that these investors do not adjust their portfolios in reaction to shifts in market conditions, potentially missing opportunities to enhance returns during favorable periods or to mitigate losses during downturns.

In contrast, the other two investors, who are aware of the regime-switching characteristics of the market (i.e., TVRA-MMV(MC) and TVRA-MMV(HMM)), exhibit markedly different allocation behaviors. The “oracle” investor (TVRA-MMV(MC)), possessing perfect knowledge of the current market state, reacts immediately after a regime change. This immediate adjustment is depicted by the sharp movements of the purple line in both Figures 3(c) and 3(d). As market conditions improve (transitioning from “bear” to “bull”), the allocation to the risky asset (S&P 500) increases accordingly.

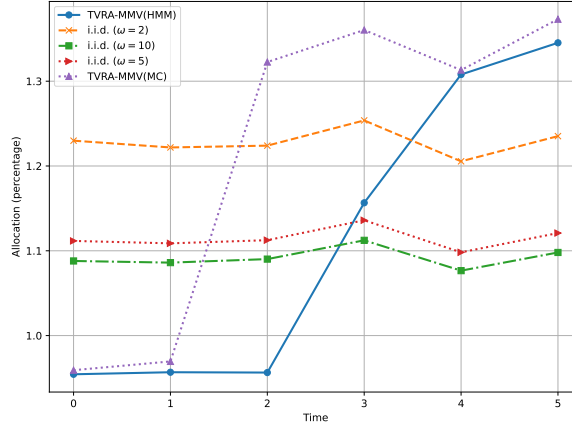
Our proposed strategy (TVRA-MMV(HMM)), on the other hand, adjusts allocations with slightly lagged response to the market shifts, shown as the blue line in both Figures 3(c) and 3(d). Since the actual state transitions are not directly observable, the investor estimates the current state and predicts future states based on observable market indicators (e.g. VIX index). Within the investment horizon, the probability of different market states may change as new data comes in. Moreover, the online estimation and implementation procedure makes it possible for risk aversion parameters to change over time. The blue line in Figure 3(b), which depicts the risk aversion parameters of TVRA-MMV(HMM) over time, also reveal this characteristic. Consequently, the policy adjustments evolve gradually, converging toward the allocations of the “oracle” investor as more information becomes available. Our proposed strategy demonstrates that even without perfect knowledge of the market state, investors can still benefit from modeling regime dynamics by utilizing observable proxies to estimate state probabilities.



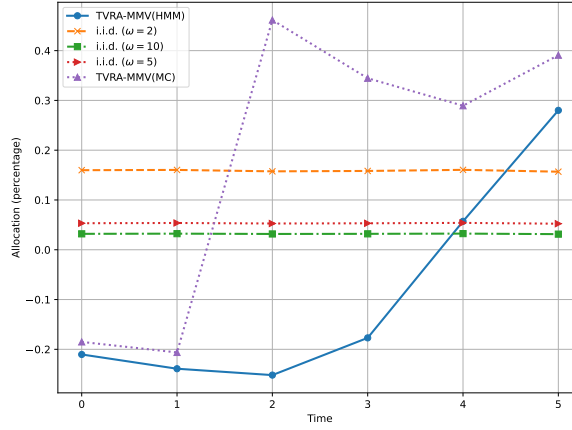
(a) State transition.



(b) Risk aversion changes.



(c) Optimal allocation when another asset is risky.



(d) Optimal allocation when another asset is risk-free.

Fig. 3: Market state transitions, time-varying risk aversions and corresponding investment strategies: (a) Sample path of market state transitions over the investment horizon; (b) Evolution of risk aversion parameters for different investment strategies; (c) Optimal allocation of S&P 500 when the other asset is DAX (risky asset); (d) Optimal allocation of S&P 500 when the other asset is T-bill (risk-free asset).

5.3 Performance Measure

To illustrate the advantage of our model that it can accommodate multiple risky assets (more than 2), the experiments in this subsection are conducted on two datasets⁷: the Fama-French 5-industry monthly return data and the Fama-French 12-industry monthly return data, spanning from January 2000 to December 2023. To evaluate the performance of our model and strategy, we consider the following return-based measures: mean, standard deviation, Sharpe ratio (SR) and adjusted Sharpe ratio (ASR). The Sharpe ratio is given by

$$SR = \frac{\mu}{\sigma},$$

where μ and σ are the mean and standard deviation of portfolio excess returns [16], respectively. The adjusted Sharpe ratio [57] can be calculated as

$$ASR = SR(1 + \frac{S}{6}SR - \frac{(K-3)}{24}SR^2),$$

where S and K represents skewness and kurtosis of the distribution with respect to returns, respectively.

We introduce a performance metric called gap to measure how close each strategy comes to the oracle's performance. The gap is calculated as:

$$gap = \left(1 - \frac{SR}{SR_{\text{oracle}}}\right) \times 100\%, \quad (38)$$

where SR is the Sharpe ratio of the strategy being evaluated (i.i.d. ($\omega = 2$), i.i.d. ($\omega = 5$), i.i.d. ($\omega = 10$) or TVRA-MMV(HMM)) and SR_{oracle} is the Sharpe ratio of TVRA-MMV(MC). Since the oracle knows the true state of the Markov chain model, its Sharpe ratio will outperform that of other models. In this context, the gap value measures the performance loss resulting from model misspecification or lack of state informaion. Therefore, a smaller gap percentage indicates better performance, with 0% signifying performance equal to that of the oracle.

Table 1 and 2 summarize the out-of-sample investment performance of optimal portfolio strategies with respect to various models. Our model (TVRA-MMV(HMM)) are compared with other models, including i.i.d. ($\omega = 2$), i.i.d. ($\omega = 5$) and i.i.d. ($\omega = 10$), where the models' settings are identically to those described in Section 5.2.

In the situation where investment in the risk-free asset is allowed, the statistics in Table 1 and 2 reveals that the most aggressive investor with a low risk aversion $\omega = 2$ achieves higher mean returns and volatility compared to the most conservative investor with a high risk aversion coefficient $\omega = 10$. The investor with $\omega = 5$ avoids potential loss compared to the aggressive investor (lower volatility) and gain more profit compared the conservative investor (higher return). However, none of them

⁷The Fama-French 5-industry portfolios monthly returns and Fama-French 12-industry portfolios monthly returns can be found from the Kenneth R.French website <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/>.

Table 1 Performance Metrics for Different Models on 5 Industry Portfolio over 6 Periods (with and without risk-free asset). Note: Std: Standard deviation; SR: Sharpe Ratio; ASR: Adjusted Sharpe Ratio.

Model	Mean	Std	SR	Gap	ASR
With Risk-Free Asset					
i.i.d. ($\omega = 2$)	1.0758	0.0974	0.6835	22.91%	0.6322
i.i.d. ($\omega = 10$)	1.0224	0.0194	0.6833	22.93%	0.6317
i.i.d. ($\omega = 5$)	1.0331	0.0346	0.6922	21.93%	0.6471
TVRA-MMV(HMM)	1.0876	0.0930	0.8438	4.83%	0.9346
Without Risk-Free Asset					
i.i.d. ($\omega = 2$)	1.0419	0.0919	0.4561	32.10%	0.4598
i.i.d. ($\omega = 10$)	1.0463	0.0788	0.5875	12.54%	0.6008
i.i.d. ($\omega = 5$)	1.0452	0.0805	0.5611	16.47%	0.5740
TVRA-MMV(HMM)	1.0520	0.0819	0.6352	5.43%	0.6282

Table 2 Performance Metrics for Different Models on 12 Industry Portfolio over 6 Periods (with and without risk-free asset). Note: Std: Standard deviation; SR: Sharpe Ratio; ASR: Adjusted Sharpe Ratio.

Model	Mean	Std	SR	Gap	ASR
With Risk-Free Asset					
i.i.d. ($\omega = 2$)	1.1495	0.1685	0.8323	44.37%	0.7616
i.i.d. ($\omega = 10$)	1.0372	0.0336	0.8324	44.36%	0.7616
i.i.d. ($\omega = 5$)	1.0579	0.0593	0.8210	45.13%	0.7591
TVRA-MMV(HMM)	1.1165	0.1062	1.0105	32.46%	1.1773
Without Risk-Free Asset					
i.i.d. ($\omega = 2$)	1.0973	0.1955	0.4976	54.44%	0.4414
i.i.d. ($\omega = 10$)	1.0620	0.0887	0.6993	35.98%	0.6041
i.i.d. ($\omega = 5$)	1.0708	0.1085	0.6530	40.22%	0.5619
TVRA-MMV(HMM)	1.0972	0.1064	0.9133	16.49%	0.7936

achieve the best performance in terms of Sharpe ratio or adjusted Sharpe ratio since they fail to adequately capture the dynamics of market state changes, which has been analysed in the experiments displayed in Section 5.2.

Our TVRA-MMV(HMM) model achieves balanced performance with moderate mean returns and volatility, while demonstrating superior performance metrics - particularly in Sharpe ratio, gap, and adjusted Sharpe ratio - compared to the naive i.i.d. models with fixed risk aversion coefficients.

The model’s effectiveness stems from its ability to adapt to market conditions, as shown in Section 5.2. It successfully differentiates between market states, increasing aggression during bull markets and exercising caution during bear markets in a manner similar to the TVRA-MMV(MC) model. This adaptive behavior results in the smallest performance gap among compared models. While the HMM-based approach may experience some delay in responding to abrupt market changes due to its reliance on indirect state observations (resulting in a non-zero gap compared to TVRA-MMV(MC)), the empirical results in Tables 1 and 2 demonstrate that our model maintains a clear performance advantage over naive strategies in dynamic market conditions.

6 Conclusion

To address the challenge that investors’ risk preferences are influenced by unobservable market conditions, this work presents significant advancements in MMV portfolio optimization by introducing a practical and tractable model that incorporates unobservable regime-switching dynamics and market-state-dependent risk aversion. By leveraging a hidden Markov model (HMM), we effectively estimate transition probabilities and filter market states, enabling robust portfolio decision-making. Closed-form portfolio policies are derived for two key scenarios: one involving only risky assets and the other including a risk-free asset. These policies overcome the challenges posed by unobservable parameters and generalize existing MMV models under full market information. A key feature of our approach is its adaptability through online implementation, which dynamically utilizes the most recent data while discarding outdated information, enhancing its real-world applicability. Empirical validation using real market data demonstrates that our model consistently outperforms benchmark strategies, underscoring the benefits of incorporating regime-switching dynamics and time-varying risk aversion. These findings highlight the risks of oversimplification in portfolio optimization and emphasize the importance of sophisticated modeling to achieve superior performance in dynamic financial markets.

This study makes a significant contribution to the field of MMV portfolio optimization by introducing dynamic risk aversion. Our methodology, which leverages the hidden Markov chain to model the stochastic nature of market states, provides a robust framework for investors to navigate varying market conditions. The inclusion of a time-varying risk aversion coefficient allows for a more nuanced investment strategy that adapts to prevailing market environments, aligning with behavioral patterns observed in real-world investment decisions.

For future research, this framework could be expanded to incorporate more realistic trading constraints, such as transaction costs or regulatory restrictions, further enhancing its practical utility in the ever-changing financial markets. Additionally,

beyond the HMM-based learning model, our framework could be extended to more general learning and optimization approaches, such as dynamic factor models or machine learning techniques, to further refine portfolio strategies and address the complexities of modern financial markets.

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Declarations

Non-financial interests: The authors have no competing interests to declare that are relevant to the content of this article.

Data availability statement: All industry portfolio data were obtained from Fama and French Data library at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Appendix A Proof in Section 4.3.1

A.1 Proof of Lemma 1

Proof. In Algorithm 3, by the definition of $\tilde{\gamma}_k(i)$ and $\mathbf{D}_k(i)$, since $\mathbf{D}_k(i)$ is positive definite for all $k = t, \dots, T$, $\mathbf{1}'\bar{\mathbf{D}}_k^{-1}\mathbf{D}_k(i)\bar{\mathbf{D}}_k^{-1}\mathbf{1} > 0$ always hold. When $k = T$, $\tilde{\gamma}_T(i) = 1$ and $\hat{\gamma}_T(i) = 1$, it follows that $\tilde{\gamma}_{T-1}(i) > 0$ since $\tilde{\gamma}_{T-1}(i) = \hat{\gamma}_T(i)d_T^2\mathbf{1}'\bar{\mathbf{D}}_T^{-1}\mathbf{D}_T(i)\bar{\mathbf{D}}_T^{-1}\mathbf{1} > 0$ and $\hat{\gamma}_{T-1}(i) > 0$ by the relationship $\hat{\gamma}_{T-1}(i) = \sum_{j \in \mathbb{M}} Q_t(i, j)\tilde{\gamma}_{T-1}(j)$. By mathematical induction, if at time $k+1$ ($k \geq t$), $\tilde{\gamma}_{k+1}(i) > 0$ and $\hat{\gamma}_{k+1}(i) > 0$ for all $i \in \mathbb{M}$, we can prove that $\tilde{\gamma}_k(i) > 0$ and $\hat{\gamma}_k(i) > 0$ always hold in a similar way.

By the definition of $\bar{\mathbf{D}}_t$ in Algorithm 3, it can be viewed as a weighted combination of $\mathbf{D}_t(i)$ whose weights are positive, thus, $\bar{\mathbf{D}}_t$ is positive definite. Therefore, $\mathbf{1}'\bar{\mathbf{D}}_t^{-1}\mathbf{1} > 0$ and $a_t > 0$. $\square$

A.2 Proof of Theorem 1

Proof. Let $\{\mathbf{u}_k\}_{k=t}^{T-1} \triangleq (\mathbf{u}_t, \mathbf{u}_{t+1}, \dots, \mathbf{u}_{T-1})$ be a policy of the truncated problem $\mathcal{A}_{all-risky}(\epsilon_t, \omega_t)|_t^T$ given filtration $\mathcal{F}_t$. We additionally introduce $f_t(i, x_t)$ as the value function where at starting time t the market regime is $s_t = i$ and the amount of money available for investment is x_t , and it can be written as

$$f_t(i, x_t) = \mathbb{E}_t \left[-\omega_t x_T^2 + \epsilon_t x_T | s_t = i \right], \quad (\text{A1})$$

where x_T is the terminal wealth when the strategy $\{\mathbf{u}_k\}_{k=t}^{T-1}$ is implemented. In other words, $f_t(i, x_t)$ is obtained by separating the value function $v_t(x_t)$ by market state s_t .

We use the condition $s_t = i$ to emphasize its dependency on the state i . Using the properties of conditional expectation, we can express the value function as

$$v_t(x_t) = \max_{\mathbf{u}_t} \sum_{i \in \mathbb{M}} \mathbb{P}(s_t = i) f_t(i, x_t). \quad (\text{A2})$$

Moreover, by the law of iterated expectations [58], we have

$$f_k(i, x_k) = \mathbb{E}_k[f_{k+1}(s_{k+1}, x_{k+1}) | s_k = i] \quad (\text{A3})$$

for $k \geq t$, with the terminal condition $f_T(i, x_T) = -\omega_T x_T^2 + \epsilon_T x_T$ for all $i \in \mathbb{M}$. Note that the difference between $f_k(i, x_k)$ and $v_k(x_k)$ is that for $v_k(x_k)$, we only know the wealth level is x_k , while for $f_k(i, x_k)$, the state i is fixed and known as a prior. Since we adopt a hidden Markov model to describe the market states, at each time, what we actually know is merely the wealth level. However, we can separately compute the value function $f_k(i, x_k)$ by assuming the market state s_k .

Let $l_k = \bar{\alpha}_k + \bar{\mathbf{r}}_k^\top \bar{\mathbf{D}}_k^{-1} \bar{\mathbf{r}}_k - \frac{(\mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \bar{\mathbf{r}}_k)^2}{\mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{1}}$ for $k \in \{t, \dots, T-1\}$ and $l_T = 0$. We claim that both value functions take quadratic forms:

$$\begin{aligned} v_k(x_k) &= -\omega_t d_k x_k^2 + \epsilon_t j_k x_k + \frac{\epsilon_t^2}{4\omega_t} l_k, \\ f_k(i, x_k) &= -\omega_t \tilde{\gamma}_k(i) x_k^2 + \epsilon_t \tilde{\rho}_k(i) x_k + \frac{\epsilon_t^2}{4\omega_t} \tilde{\alpha}_k(i). \end{aligned} \quad (\text{A4})$$

We can prove this claim by backward induction. At stage $k = T$, since $d_T = 1$, $j_T = 1$, $l_T = 0$ and $\tilde{\gamma}_T(i) = 1$, $\tilde{\rho}_T(i) = 1$, $\tilde{\alpha}_T(i) = 0$ for all $i \in \mathbb{M}$, the two claims are true automatically. For any $k \geq t$, assume that the claims are true at stage $k+1$, we prove it also holds at time k . At stage k , the value function $v_k(x_k)$ can be written as follows:

$$\begin{aligned} v_k(x_k) &= \max_{x_k = \mathbf{1}^\top \mathbf{u}_k} \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) f_k(i, x_k) \\ &= \max_{x_k = \mathbf{1}^\top \mathbf{u}_k} \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \\ &\quad \mathbb{E}_k \left[-\omega_t \tilde{\gamma}_{k+1}(s_{k+1}) x_{k+1}^2 + \epsilon_t \tilde{\rho}_{k+1}(s_{k+1}) x_{k+1} + \frac{\epsilon_t^2}{4\omega_t} \tilde{\alpha}_{k+1}(s_{k+1}) \right] \\ &= \max_{x_k = \mathbf{1}^\top \mathbf{u}_k} \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \sum_{j \in \mathbb{M}} Q_t(i, j) \\ &\quad \mathbb{E}_k \left[-\omega_t \tilde{\gamma}_{k+1}(j) x_{k+1}^2 + \epsilon_t \tilde{\rho}_{k+1}(j) x_{k+1} + \frac{\epsilon_t^2}{4\omega_t} \tilde{\alpha}_{k+1}(j) \right] \\ &= \max_{x_k = \mathbf{1}^\top \mathbf{u}_k} \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \sum_{j \in \mathbb{M}} Q_t(i, j) \\ &\quad \mathbb{E}_k \left[-\omega_t \tilde{\gamma}_{k+1}(j) (\mathbf{R}_{k+1}^\top \mathbf{u}_k)^2 + \epsilon_t \tilde{\rho}_{k+1}(j) \mathbf{R}_{k+1}^\top \mathbf{u}_k + \frac{\epsilon_t^2}{4\omega_t} \tilde{\alpha}_{k+1}(j) \right] \end{aligned}$$

$$= \max_{x_k = \mathbf{1}^\top \mathbf{u}_k} -\omega_t \mathbf{u}_k^\top \bar{\mathbf{D}}_k \mathbf{u}_k + \epsilon_t \bar{\mathbf{r}}_k^\top \mathbf{u}_k + \frac{\epsilon_t^2}{4\omega_t} \bar{\alpha}_k.$$

To solve this problem, we consider the following Lagrangian function by introducing a multiplier d ,

$$L(d) = -\omega_t \mathbf{u}_k^\top \bar{\mathbf{D}}_k \mathbf{u}_k + \epsilon_t \bar{\mathbf{r}}_k^\top \mathbf{u}_k + d(x_k - \mathbf{1}^\top \mathbf{u}_k). \quad (\text{A5})$$

Since we have proved $\bar{\mathbf{D}}_k$ is a positive definite matrix in Lemma 1, this is a concave function with respect to $\mathbf{u}_k$, and consequently the optimal solution must satisfy the first-order optimal condition, i.e.,

$$\nabla_{\mathbf{u}_k} L(d) = 2\omega_t \bar{\mathbf{D}}_k \mathbf{u}_k - \epsilon_t \bar{\mathbf{r}}_k - d\mathbf{1} = 0. \quad (\text{A6})$$

In other words, the optimal $\mathbf{u}_k^*$ as well as dual variable d^* should satisfy

$$\begin{aligned} \mathbf{u}_k^* &= \frac{1}{2\omega_t} \bar{\mathbf{D}}_k^{-1} (\epsilon_t \bar{\mathbf{r}}_k + d^* \mathbf{1}), \\ d^* &= \frac{2\omega_t x_k - \epsilon_t \mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \bar{\mathbf{r}}_k}{\mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{1}}. \end{aligned} \quad (\text{A7})$$

By substituting $\mathbf{u}_k^*$ into $v_k(x_k)$ and $f_k(i, x_k)$, the value functions become

$$\begin{aligned} v_k(x_k) &= -\frac{1}{4\omega_t} [(\epsilon_t \bar{\mathbf{r}}_k + d^* \mathbf{1})^\top \bar{\mathbf{D}}_k^{-1} (\epsilon_t \bar{\mathbf{r}}_k + d^* \mathbf{1})] + \frac{\epsilon_t}{2\omega_t} \bar{\mathbf{r}}_k^\top \bar{\mathbf{D}}_k^{-1} (\epsilon_t \bar{\mathbf{r}}_k + d^* \mathbf{1}) + \frac{\epsilon_t^2}{4\omega_t} \bar{\alpha}_k \\ &= -\frac{(d^*)^2}{4\omega_t} \mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{1} + \frac{\epsilon_t^2}{4\omega_t} (\bar{\mathbf{r}}_k^\top \bar{\mathbf{D}}_k^{-1} \bar{\mathbf{r}}_k + \bar{\alpha}_k) \\ &= -\omega_t \frac{1}{\mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{1}} x_k^2 + \epsilon_t \frac{\mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \bar{\mathbf{r}}_k}{\mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{1}} x_k + \frac{\epsilon_t^2}{4\omega_t} [\bar{\alpha}_{k+1} + \bar{\mathbf{r}}_k^\top \bar{\mathbf{D}}_k^{-1} \bar{\mathbf{r}}_k - \frac{(\mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \bar{\mathbf{r}}_k)^2}{\mathbf{1}^\top \bar{\mathbf{D}}_k^{-1} \mathbf{1}}] \\ &= -\omega_t d_k x_k^2 + \epsilon_t j_k x_k + \frac{\epsilon_t^2}{4\omega_t} l_k, \end{aligned}$$

and

$$\begin{aligned} f_k(i, x_k) &= \mathbb{E}_k \left[-\omega_t \tilde{\gamma}_{k+1}(s_{k+1}) x_{k+1}^2 + \epsilon_t \tilde{\rho}_{k+1}(s_{k+1}) x_{k+1} + \frac{\epsilon_t^2}{4\omega_t} \tilde{\alpha}_{k+1}(s_{k+1}) | s_k = i \right] \\ &= \sum_{j \in \mathbb{M}} Q_t(i, j) \mathbb{E}_k \left[-\omega_t \tilde{\gamma}_{k+1}(j) x_{k+1}^2 + \epsilon_t \tilde{\rho}_{k+1}(j) x_{k+1} + \frac{\epsilon_t^2}{4\omega_t} \tilde{\alpha}_{k+1}(j) | s_k = i \right] \\ &= \sum_{j \in \mathbb{M}} Q_t(i, j) \\ &\quad \mathbb{E}_k [-\omega_t \tilde{\gamma}_{k+1}(j) (\mathbf{R}_{k+1}^\top \mathbf{u}_k^*)^2 + \epsilon_t \tilde{\rho}_{k+1}(j) \mathbf{R}_{k+1}^\top \mathbf{u}_k^* + \frac{\epsilon_t^2}{4\omega_t} \tilde{\alpha}_{k+1}(j) | s_k = i] \\ &= -\omega_t (\mathbf{u}_k^*)^\top \hat{\gamma}_{k+1}(i) \mathbf{D}_k(i) \mathbf{u}_k^* + \epsilon_t \hat{\rho}_{k+1}(i) (\mathbf{r}_k(i))^\top \mathbf{u}_k^* + \frac{\epsilon_t^2}{4\omega_t} \hat{\alpha}_{k+1}(i) \end{aligned}$$

$$= -\omega_t \tilde{\gamma}_k(i) x_k^2 + \epsilon_t \tilde{\rho}_k(i) x_k + \frac{\epsilon_t^2}{4\omega_t} \tilde{\alpha}_k(i).$$

The corresponding optimal policy to $\mathcal{A}_{all-risky}(\epsilon_t, \omega_t)$ is given by

$$\mathbf{u}_k^*(x_k, \epsilon_t) = \frac{\epsilon_t}{2\omega_t} \bar{\mathbf{D}}_k^{-1} (\bar{\mathbf{r}}_k - j_k \mathbf{1}) + d_k \bar{\mathbf{D}}_k^{-1} \mathbf{1} x_k \quad \forall k \geq t. \quad (\text{A8})$$

Therefore, at time t , it has

$$\begin{aligned} \mathbf{u}_t^*(x_t, \epsilon_t) &= \frac{\epsilon_t}{2\omega_t} \bar{\mathbf{D}}_t^{-1} (\bar{\mathbf{r}}_t - b_t \mathbf{1}) + a_t \bar{\mathbf{D}}_t^{-1} \mathbf{1} x_t, \\ v_t(x_t) &= -\omega_t a_t x_t^2 + \epsilon_t b_t x_t + \frac{\epsilon_t^2}{4\omega_t} c_t. \end{aligned} \quad (\text{A9})$$

□

A.3 Proof of Theorem 2

Proof. Given $\mathcal{F}_t$, at time t the wealth is x_t , by taking the optimal policy $\mathbf{u}_t^*(x_t, \epsilon_t)$ into the wealth process, we have

$$\begin{aligned} x_{t+1} &= \mathbf{R}_{t+1}^\top \mathbf{u}_t^* \\ &= \frac{\epsilon_t}{2\omega_t} \mathbf{R}_{t+1}^\top \bar{\mathbf{D}}_t^{-1} (\bar{\mathbf{r}}_t - b_t \mathbf{1}) + a_t \mathbf{R}_{t+1}^\top \bar{\mathbf{D}}_t^{-1} \mathbf{1} x_t. \end{aligned} \quad (\text{A10})$$

Assume the market state is $s_t = i$, multiplying $\hat{\rho}_{t+1}(i)$ and then taking expectation on both sides gives rise to the following equation

$$\mathbb{E}_t[\hat{\rho}_{t+1}(i) x_{t+1} | s_t = i] = \frac{\epsilon_t}{2\omega_t} \hat{\rho}_{t+1}(i) \mathbf{r}_t^\top(i) \bar{\mathbf{D}}_t^{-1} (\bar{\mathbf{r}}_t - b_t \mathbf{1}) + a_t \hat{\rho}_{t+1}(i) \mathbf{r}_t^\top(i) \bar{\mathbf{D}}_t^{-1} \mathbf{1} x_t. \quad (\text{A11})$$

Taking summation over all possibility with respect to market state $\mathbb{P}(s_t = i | \mathcal{F}_t)$ and using the law of iterated expectation with the boundary condition $\hat{\rho}_T(i) = 1$ for all $i \in \mathbb{M}$, the first result in Theorem 2 is straightforward. We can prove the second equation in a similar approach. Taking square on both sides of (A10), we have

$$\begin{aligned} x_{t+1}^2 &= (\mathbf{R}_{t+1}^\top \mathbf{u}_t^*)^2 \\ &= \left(\frac{\epsilon_t}{2\omega_t} \right)^2 (\bar{\mathbf{r}}_t - b_t \mathbf{1})^\top \bar{\mathbf{D}}_t^{-1} \mathbf{R}_{t+1} \mathbf{R}_{t+1}^\top \bar{\mathbf{D}}_t^{-1} (\bar{\mathbf{r}}_t - b_t \mathbf{1}) \\ &\quad + a_t^2 \mathbf{1}^\top \bar{\mathbf{D}}_t^{-1} \mathbf{R}_{t+1} \mathbf{R}_{t+1}^\top \bar{\mathbf{D}}_t^{-1} \mathbf{1} x_t^2 \\ &\quad + \frac{\epsilon_t}{\omega_t} a_t \mathbf{1}^\top \bar{\mathbf{D}}_t^{-1} \mathbf{R}_{t+1} \mathbf{R}_{t+1}^\top \bar{\mathbf{D}}_t^{-1} (\bar{\mathbf{r}}_t - b_t \mathbf{1}) x_t. \end{aligned} \quad (\text{A12})$$

Assume the initial market state is $s_t = i$, multiplying $\hat{\gamma}_{t+1}(i)$ and taking expectation on both sides, we have

$$\begin{aligned}
& \mathbb{E}_t[\hat{\gamma}_{t+1}(i)x_{t+1}^2 | s_t = i] \\
&= \left(\frac{\epsilon_t}{2\omega_t}\right)^2 \hat{\gamma}_{t+1}(i)(\bar{\mathbf{r}}_t - b_t \mathbf{1})^\top \bar{\mathbf{D}}_t^{-1} \mathbf{D}(i) \bar{\mathbf{D}}_t^{-1} (\bar{\mathbf{r}}_t - b_t \mathbf{1}) \\
&+ a_t^2 \hat{\gamma}_{t+1}(i) \mathbf{1}^\top \bar{\mathbf{D}}_t^{-1} \mathbf{D}(i) \bar{\mathbf{D}}_t^{-1} \mathbf{1} x_t^2 \\
&+ \frac{\epsilon_t}{\omega_t} a_t \hat{\gamma}_{t+1}(i) \mathbf{1}^\top \bar{\mathbf{D}}_t^{-1} \mathbf{D}(i) \bar{\mathbf{D}}_t^{-1} (\bar{\mathbf{r}}_t - b_t \mathbf{1}) x_t,
\end{aligned} \tag{A13}$$

then taking summation over all possibility of market state s_t and using the boundary condition $\hat{\gamma}_T(i) = 1$ for all $i \in \mathbb{M}$ gives the second equation. $\square$

A.4 Proof of Lemma 2

Proof. To prove $0 < c_t < 1$, note that

$$\text{Var}_t[X_T] = (a_t - b_t^2)x_t^2 - \frac{\epsilon_t}{\omega_t} b_t c_t x_t + \left(\frac{\epsilon_t}{2\omega_t}\right)^2 (1 - c_t) c_t,$$

when x_t is set to zero, the variance becomes

$$\text{Var}_t[X_T] = \left(\frac{\epsilon_t}{2\omega_t}\right)^2 (1 - c_t) c_t.$$

However, the risky asset is non-degenerate, thus the risk of the whole portfolio still exists and can't be completely hedged. Therefore, $(1 - c_t)c_t > 0$ must hold, which implies $0 < c_t < 1$. $\square$

Appendix B Proof in Section 4.3.2

B.1 Proof of Theorem 4

Proof. This proof of Theorem 4 is similar to that of Theorem 1. Let $\{\mathbf{u}_k\}_{k=t}^{T-1} \triangleq (\mathbf{u}_t, \mathbf{u}_{t+1}, \dots, \mathbf{u}_{T-1})$ be a policy of the truncated problem $\mathcal{A}_{risk-free}(\lambda_t, \omega_t)|_t^T$. We additionally introduce $\phi_t(i, x_t)$ as the value function where at starting time t the market regime is $s_t = i$ and the amount of money available for investment is x_t , and it can be written as

$$\phi_t(i, x_t) = \mathbb{E}_t[-\omega_t x_T^2 + \lambda_t x_T | s_t = i], \tag{B14}$$

where x_T is the terminal wealth when the strategy $\{\mathbf{u}_k\}_{k=t}^{T-1}$ is implemented. $\phi_t(i, x_t)$ is obtained by separating the value function $\kappa_t(x_t)$ by market state s_t . We can show that

$$\kappa_t(x_t) = \max_{\mathbf{u}_t} \sum_{i \in \mathbb{M}} \mathbb{P}(s_t = i) \phi_t(i, x_t), \tag{B15}$$

with the recursive relationship

$$\phi_k(i, x_k) = \mathbb{E}_k[\phi_{k+1}(s_{k+1}, x_{k+1}) | s_k = i], \quad (\text{B16})$$

for $k \geq t$, with the terminal condition $\phi_T(i, x_T) = -\omega_t x_T^2 + \lambda_t x_T$ for all $i \in \mathbb{M}$. We will prove the recursion is true by backward induction starting with the boundary conditions. At time T , the claim is true in that $\tilde{\omega}_T(i) = 1, \tilde{\lambda}_T(i) = 1, \tilde{\eta}_T(i) = 0$ for all $i \in \mathbb{M}$. Suppose now that the induction hypothesis holds for $k+1$ ($k > t$), then at stage k , the function $\phi_k(i, x_k)$ becomes

$$\begin{aligned} \phi_k(i, x_k) &= \sum_{j \in \mathbb{M}} Q_t(i, j) \mathbb{E}_k \left[\phi_{k+1} \left(j, r_{f,k}(i) x_k + \mathbf{R}_{k+1}^e(i)^\top \mathbf{u}_k \right) \right] \\ &= \sum_{j \in \mathbb{M}} Q_t(i, j) \mathbb{E}_k \left[-\omega_t \cdot \tilde{\omega}_{k+1}(j) \left(r_{f,k}(i) x_k + \mathbf{R}_{k+1}^e(i)^\top \mathbf{u}_k \right)^2 \right. \\ &\quad \left. + \lambda_t \cdot \tilde{\lambda}_{k+1}(j) \left(r_{f,k}(i) x_k + \mathbf{R}_{k+1}^e(i)^\top \mathbf{u}_k \right) + \frac{\lambda_t^2}{4\omega_t} \cdot \tilde{\eta}_{k+1}(j) \right]. \quad (\text{B17}) \\ &= -\omega_t \cdot \hat{\omega}_{k+1}(i) r_{f,k}(i)^2 x_k^2 + \lambda_t \cdot \hat{\lambda}_{k+1}(i) r_{f,k}(i) x_k \\ &\quad + \left(\lambda_t \cdot \hat{\lambda}_{k+1}(i) - 2\omega_t \cdot \hat{\omega}_{k+1}(i) r_{f,k}(i) x_k \right) \mathbf{r}_k^e(i)^\top \mathbf{u}_k \\ &\quad - \omega_t \cdot \hat{\omega}_{k+1}(i) \mathbf{u}_k^\top \mathbf{Z}_k(i) \mathbf{u}_k + \frac{\lambda_t^2}{4\omega_t} \hat{\eta}_{k+1}(i). \end{aligned}$$

As a consequence, $\kappa_k(x_k)$ can be rewritten as

$$\begin{aligned} \kappa_k(x_k) &= \max_{\mathbf{u}_k} \sum_{i \in \mathbb{M}} \mathbb{P}(s_k = i | \mathcal{F}_t) \phi_k(i, x_k) \\ &= \max_{\mathbf{u}_k} -\omega_t \bar{r}_{2f,k} x_k^2 + \lambda_t \bar{r}_{f,k} x_k + \mathbf{u}_k^\top (\lambda_t \bar{\mathbf{r}}_{e,k} - 2\omega_t x_k \bar{\mathbf{r}}_{ef,k}) \\ &\quad - \omega_t \mathbf{u}_k^\top \bar{\mathbf{Z}}_k \mathbf{u}_k + \frac{\lambda_t^2}{4\omega_t} \bar{\eta}_{k+1}. \quad (\text{B18}) \end{aligned}$$

The same argument can be repeated to find the optimal solution

$$\mathbf{u}_k^*(x_k, \lambda_t) = \frac{1}{2} \left(\frac{\lambda_t}{\omega_t} \right) \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{e,k} - \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} x_k \quad (\text{B19})$$

and substituting this optimal solution into the value functions $\kappa_k(x_k)$ and $\phi_k(i, x_k)$ gives the expressions for parameters

$$\begin{aligned} h_k &= \bar{r}_{2f,k} - \bar{\mathbf{r}}_{ef,k}^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k}, \\ p_k &= \bar{r}_{f,k} - \bar{\mathbf{r}}_{ef,k}^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{e,k}, \\ q_k &= \bar{\mathbf{r}}_{e,k}^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{e,k} + \bar{\eta}_k, \\ \tilde{\omega}_k(i) &= \hat{\omega}_{k+1}(i) r_{f,k}(i)^2 - 2\hat{\omega}_{k+1}(i) r_{f,k}(i) \mathbf{r}_k^e(i)^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} \end{aligned}$$

$$\begin{aligned}
& + \hat{\omega}_{k+1}(i) \bar{\mathbf{r}}_{ef,k}^\top \bar{\mathbf{Z}}_k^{-1} \mathbf{Z}_k(i) \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k}, \\
\tilde{\lambda}_k(i) &= \hat{\lambda}_{k+1}(i) r_{f,k}(i) - \hat{\lambda}_{k+1}(i) \mathbf{r}_k^e(i)^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} \\
& - \hat{\omega}_{k+1}(i) r_{f,k}(i) \mathbf{r}_k^e(i)^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} \\
& + \hat{\omega}_{k+1}(i) \bar{\mathbf{r}}_{ef,k}^\top \bar{\mathbf{Z}}_k^{-1} \mathbf{Z}_k(i) \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k}, \\
\tilde{\eta}_k(i) &= 2\hat{\lambda}_{k+1}(i) \mathbf{r}_k^e(i)^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} - \hat{\omega}_{k+1}(i) \bar{\mathbf{r}}_{ef,k}^\top \bar{\mathbf{Z}}_k^{-1} \mathbf{Z}_k(i) \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} + \hat{\eta}_{k+1}(i).
\end{aligned}$$

Therefore, at time t , the optimal allocation is given by

$$\mathbf{u}_t^*(x_t, \lambda_t) = \frac{1}{2} \left(\frac{\lambda_t}{\omega_t} \right) \bar{\mathbf{Z}}_t^{-1} \bar{\mathbf{r}}_{e,t} - \bar{\mathbf{Z}}_t^{-1} \bar{\mathbf{r}}_{ef,t} x_t, \quad (\text{B20})$$

along with the value function

$$\kappa_t(x_t) = -\omega_t h_t x_t^2 + \lambda_t p_t x_t + \frac{\lambda_t^2}{4\omega_t} q_t. \quad (\text{B21})$$

□

B.2 Proof of Lemma 5

Lemma 5. *For any random variable ζ_i, δ_i, ξ_i that are non-degenerate,*

$$\sum_{i \in \mathbb{M}} \zeta_i \delta_i^2 \sum_{i \in \mathbb{M}} \zeta_i \mathbb{E}(\xi_i^2) > \left(\sum_{i \in \mathbb{M}} \zeta_i \delta_i \mathbb{E}(\xi_i) \right)^2. \quad (\text{B22})$$

Proof. Note that for any random variable that is non-degenerate,

$$\begin{aligned}
& \left(\sum_{i \in \mathbb{M}} \zeta_i \delta_i^2 \right) \left(\sum_{i \in \mathbb{M}} \zeta_i \mathbb{E}(\xi_i^2) \right) \\
&= \left(\sum_{i \in \mathbb{M}} \zeta_i^2 \delta_i^2 \mathbb{E}(\xi_i^2) \right) + \frac{1}{2} \sum_{i \neq j} \zeta_i \zeta_j (\delta_j^2 \mathbb{E}(\xi_i^2) + \delta_i^2 \mathbb{E}(\xi_j^2)) \\
&= \left(\sum_{i \in \mathbb{M}} \zeta_i^2 \delta_i^2 (\mathbb{E}(\xi_i))^2 \right) + \frac{1}{2} \sum_{i \neq j} \zeta_i \zeta_j (\delta_j^2 (\mathbb{E}(\xi_i))^2 + (\delta_i^2 \mathbb{E}(\xi_j))^2) \\
&+ \sum_{i \in \mathbb{M}} \zeta_i^2 \delta_i^2 \text{Var}(\xi_i) + \frac{1}{2} \sum_{i \neq j} \zeta_i \zeta_j (\delta_j^2 \text{Var}(\xi_i) + \delta_i^2 \text{Var}(\xi_j)) \\
&> \left(\sum_{i \in \mathbb{M}} \zeta_i^2 \delta_i^2 (\mathbb{E}(\xi_i))^2 \right) + \frac{1}{2} \sum_{i \neq j} \zeta_i \zeta_j (\delta_j^2 (\mathbb{E}(\xi_i))^2 + (\delta_i^2 \mathbb{E}(\xi_j))^2)
\end{aligned}$$

$$\begin{aligned}
&\geq \left(\sum_{i \in \mathbb{M}} \zeta_i^2 \delta_i^2 (\mathbb{E}(\xi_i))^2 \right) + \sum_{i \neq j} \zeta_i \zeta_j \delta_i \delta_j \mathbb{E}(\xi_i) \mathbb{E}(\xi_j) \\
&= \left(\sum_{i \in \mathbb{M}} \zeta_i \delta_i \mathbb{E}(\xi_i) \right)^2.
\end{aligned}$$

□

B.3 Proof of Lemma 3

Proof. By the relationship of $\mathbf{Z}_k(i)$ and $\mathbf{r}_k^e(i) \mathbf{r}_k^e(i)^\top$, $0 < 1 - \mathbf{r}_k^e(i)^\top \mathbf{Z}_k^{-1}(i) \mathbf{r}_k^e(i) < 1$ always holds. When $k = T$, $\tilde{\omega}_T(i) = 1$ and $\hat{\omega}_T(i) = 1$, it follows that $\tilde{\omega}_{T-1}(i) > 0$ by the relationship of $\tilde{\omega}_{T-1}(i)$ and $\hat{\omega}_{T-1}(i)$ that $\hat{\omega}_{T-1}(i) = \sum_{j \in \mathbb{M}} Q_t(i, j) \tilde{\omega}_{T-1}(j)$. By mathematical induction, if at time $k+1$, $\tilde{\omega}_{k+1}(i) > 0$ and $\hat{\omega}_{k+1}(i) > 0$ for all $i \in \mathbb{M}$ holds, we can prove that $\tilde{\omega}_k(i) > 0$ and $\hat{\omega}_k(i) > 0$ by the following inequality

$$\begin{aligned}
\tilde{\omega}_k(i) &= \hat{\omega}_{k+1}(i) r_{f,k}(i)^2 - 2\hat{\omega}_{k+1}(i) r_{f,k}(i) \mathbf{r}_k^e(i)^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} \\
&\quad + \hat{\omega}_{k+1}(i) \bar{\mathbf{r}}_{ef,k}^\top \bar{\mathbf{Z}}_k^{-1} \mathbf{Z}_k(i) \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} \\
&\geq -\hat{\omega}_{k+1}(i) \bar{\mathbf{r}}_{ef,k}^\top \bar{\mathbf{Z}}_k^{-1} \mathbf{r}_k^e(i) \mathbf{r}_k^e(i)^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} \\
&\quad + \hat{\omega}_{k+1}(i) \bar{\mathbf{r}}_{ef,k}^\top \bar{\mathbf{Z}}_k^{-1} \mathbf{Z}_k(i) \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} \\
&\geq -\hat{\omega}_{k+1}(i) \bar{\mathbf{r}}_{ef,k}^\top \bar{\mathbf{Z}}_k^{-1} \mathbf{r}_k^e(i) \mathbf{r}_k^e(i)^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} \\
&\quad + \hat{\omega}_{k+1}(i) \bar{\mathbf{r}}_{ef,k}^\top \bar{\mathbf{Z}}_k^{-1} \mathbf{Z}_k(i) \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} \geq 0.
\end{aligned}$$

The last inequality holds because covariance matrix $\Sigma_k(i)$ is positive definite for all $i \in \mathbb{M}$.

To prove $h_k > 0$, define $\zeta_{k,i} = \mathbb{P}(s_k = i | \mathcal{F}_t) \hat{\omega}_{k+1}(i)$, $\delta_{k,i} = r_{f,k}(i)$, and $\xi_{k,i} = \mathbf{R}_k^e(i)$, therefore

$$\begin{aligned}
h_k &= \bar{r}_{2f,k} - \bar{\mathbf{r}}_{ef,k}^\top \bar{\mathbf{Z}}_k^{-1} \bar{\mathbf{r}}_{ef,k} \\
&= \sum_{i \in \mathbb{M}} \zeta_{k,i} \delta_{k,i}^2 - \sum_{i \in \mathbb{M}} \zeta_{k,i} \delta_{k,i} \mathbb{E}(\xi_{k,i}) \left(\sum_{i \in \mathbb{M}} \zeta_{k,i} \mathbb{E}(\xi_{k,i}^2) \right)^{-1} \sum_{i \in \mathbb{M}} \zeta_{k,i} \delta_{k,i} \mathbb{E}(\xi_{k,i}).
\end{aligned}$$

It has $h_k > 0$ because of the inequality

$$\sum_{i \in \mathbb{M}} \zeta_{k,i} \delta_{k,i}^2 \sum_{i \in \mathbb{M}} \zeta_{k,i} \mathbb{E}(\xi_{k,i}^2) > \left(\sum_{i \in \mathbb{M}} \zeta_{k,i} \delta_{k,i} \mathbb{E}(\xi_{k,i}) \right)^2$$

stated in Lemma 5. As for the proof of $\bar{\mathbf{Z}}_t \succ 0$ and $h_t > 0$, they are similar to those of $\bar{\mathbf{D}}_t \succ 0$ and $a_t > 0$ in Lemma 1. Thus we omit them here. □

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